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University of Sunderland

Faculty of Technology
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PROM02 – MSc Dissertation
MSc Applied Data Science

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Predicting NBA Player Performance Using Machine Learning Methods

January 2025

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Abstract

The reliance on advanced analytics in sports, particularly basketball, has emphasized the need for robust and interpretable predictive models to evaluate player performance. This study applies machine learning techniques to predict Player Efficiency Rating (PER) and Win Share (WS) for NBA players, validating the models on WNBA datasets to assess cross-league robustness. Guided by the CRISP-DM methodology, the research utilized advanced player metrics and biodata from five NBA seasons (2019–2024), employing Random Forest, Support Vector Regressor (SVR), and Recurrent Neural Network (RNN) models. The study adopted the single-task and multi-task research approach in classifying the target variables. Results revealed that RNN outperformed others in predicting PER, while Random Forest and SVR had better outcomes in WS predictions. Cross-league testing highlighted the RNN model's robustness but also demonstrated league-specific dynamics impacting performance. Challenges such as data limitations in the WNBA underscore the need for more inclusive datasets. This study contributes to sports analytics by integrating advanced metrics, addressing cross-league predictive modelling, and offering actionable insights for team management, player scouting, and performance evaluation. The findings pave the way for future research, emphasizing equitable, data-driven decision-making in professional basketball and advancing machine learning applications in sports.

Contents

Declaration **2**

Abstract **3**

1 Introduction **6**

1.1 Background 6

1.2 Aims 7

1.3 Objectives 7

1.4 Research Approach 8

1.5 Structure of the Report 8

1.6 Ethical, Social, Professional, Legal and Security Considerations 9

2 Literature Review **11**

2.1 Sports Analytics and Machine Learning 11

2.2 Machine Learning and Deep Learning Techniques for Predicting Player Performance 12

2.3 Machine Learning Applications in the NBA 15

2.4 Challenges and Ethical Considerations 17

3 Practical Research Methodology **19**

3.1 Data Mining Framework 19

4 Analysis of Results **35**

4.1 Model Results 35

4.2 Discussion of Results 36

4.3 Feature Importance 37

4.4 Comparison of Results with Existing Literature 39

4.5 Model-Testing using Cross-League Validation 40

4.6 Strengths and Weaknesses Identified in the Models 41

5 Project Evaluation and Reflection **44**

5.1 Overview of Project 44

5.2 Objectives Review 44

5.3 Evaluation of Methodology and Results 46

5.4 Personal Reflection 47

5.5 Evaluation of Ethical, Legal, Social, Security and Professional Considerations 50

6 Conclusion and Recommendations **52**

6.1 Conclusion 52

6.2 Recommendations 53
7 Reference List 56
Appendix A. Project Proposal 64
Appendix B. Code Snippets 66

1 Introduction

1.1 Background

There has been a substantial increase in the use of machine learning in enhancing sport analytics in recent years, empowering teams to make informed and real-time decisions regarding player performance (Apostolou and Tjortjis, 2019). This is especially true for basketball, and specifically, the National Basketball Association (NBA) professional league, where predictive analysis has become significantly crucial, given the interaction between a player's performance and the overall success of the team (Zuccolotto, Manisera and Sandri, 2017). Basketball performance analytics has historically depended on basic measurements like points per game, assists made, steals, and total rebounds. Although these measures provide some understanding of the primary factors influencing performance, they frequently offer a limited overview of how the player directly influence the match (Wang and Fan, 2021). There has been concerted efforts recently by researchers to examine more advanced metrics of the game to effectively assess players' contributions holistically and not relying on the basic statistics. Player Efficiency Rating (PER), Total Shooting percentage (TSG%) and Win Share (WS) are metrics that have assumed more relevance in analytics, providing a more thorough understanding of the variables that make valuable contributions

Machine learning in sports analytics signifies an intriguing convergence of data science and traditional sports statistics. It allows for more precise and dynamic insights (Manner, 2016; Tichy, 2016). Predictive models have proved particularly valuable in basketball, a sport known for its high variability in player performance because of situational factors like opponent strength, injuries and individual player attributes such as age and experience. Common machine learning models—logistic regression, decision trees and neural networks—have been utilized to analyze these nonlinear relationships. They provide predictions that are often more accurate than traditional statistical methods. Sarlis and Tjortjis, (2020) illustrated the application of reinforcement learning in enhancing decision-making during basketball plays, thereby demonstrating the significance of machine learning in unravelling complex, time-dependent interactions between players.

With increasing adoption of data-driven decision-making in professional sports, predictive analytics has implications that extend beyond player evaluation. Accurate player performance prediction impacts not only team dynamics but also scouting processes, fanbase engagement and contract negotiations (Bunker and Susnjak, 2022 ; Aditya and Priya, 2024). As basketball continue to grow in global popularity, the need for robust, interpretable and equitable predictive models becomes imperative. Although these outcomes could pave the way for future studies such as the focus on cross-league predictive analytics, contributing to broader field of sports analytics and machine learning applications (Sarlis and Tjortjis, 2020).

This study seeks to develop a machine learning framework for predicting player performance, incorporating cross-league validation by testing NBA-trained models on WNBA datasets. This strategy seeks to fill research gaps in cross-league predictive analytics while improving model robustness and adaptability. By providing frameworks that encourage more efficient and effective decision-making in professional basketball, this research seeks to advance the domain of sports analytics.

1.2 Aims

The aim of this research is to develop machine learning models that can predict NBA player performance by exploring historical data and statistics, which will further enhance teams in making tactical decisions and maximizing winning contributions.

1.3 Objectives

The objectives for this research work are formulated below.

- Critically review existing literature on the impact of machine learning in predicting player performance in the NBA.
- Systematically collect, preprocess, and explore historical NBA data to identify key trends, correlations, and statistical distributions.
- Develop and implement a range of machine learning models, incorporating traditional statistics and advanced metrics for comprehensive player performance.
- Perform cross-league model testing and validation of NBA model on the WNBA datasets.

- Evaluate the predictive accuracy and robustness of the models using relevant performance metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE).
- Present a detailed report on the model outcomes, providing discussions and actionable recommendations for practical applications in sports analytics.

1.4 Research Approach

The research approach used in this study is to develop machine learning algorithms that can predict players performance in the NBA, using a single-task and multi-task approach. The collated datasets covered a period of five seasons (2019/2020 to 2023/2024) and were reliably sourced from basketball-reference.com and nba.com, which is the official website of the NBA. Data preprocessing, analysis and feature selection will be performed, setting the foundation for model development. Random Forest, Support Vector Machine and Recurrent Neural Networks (RNN) have been selected for the research and will then be developed and applied to the dataset to predict performance. Finally, the models will be evaluated with appropriate metrics to give a comparative analysis about their performance and compare the results with the project objectives.

1.5 Structure of the Report

This study adheres to the conventional framework of an academic research work and is therefore structured into six chapters, each with its varied sub-sections. These are explained further below.

Introduction: In this section, a succinct background of the study is provided, to fully understand the motivation behind the project. It also enumerates the research aim, objectives, methodology and ethical considerations. It outlines the rationale behind employing machine learning techniques in predicting NBA players performance, setting the context for subsequent exploration.

Literature Review: This section reveals the theoretical and empirical aspects of previous studies on the application of various machine learning models to player performance in the NBA. It provides critical examination of the arguments, methodologies and results of these studies, which then establishes the core foundation for this research to identify gaps.

Practical Methodology: The practical measures needed to actualise the objectives of this study effort are highlighted in this section, which also describes the techniques used in the pre-processing and collecting of data sets, feature extraction, and creation of the machine learning algorithms used to forecast performance.

Analysis of Results: In this section, the project overview will be considered, comprising the key findings, challenges faced, and insights found while implementing the various algorithms on the NBA and WNBA datasets. An evaluation of the methodology's strengths and weaknesses is included, along with recommendations for additional research initiatives.

Project Evaluation and Reflection: This section offers a thoughtful analysis and reflective examination of research procedures, encompassing personal insights: it covers the challenges, as well as lessons learnt during the study. The goal of critical self-evaluation is to improve validity and rigor of results; however, this also promotes professional and personal development during research process. Although challenges exist, they contribute to a deeper understanding. Because of this, one must embrace the complexities involved.

Conclusions and Recommendations: Lastly, the study provides a comprehensive summary of key discoveries and their implications; however, the paper offers suggestions for future research topics. It identifies areas that need more investigation and improvement in the use of machine learning to predict NBA player performance. Although the findings are significant, there remains a necessity for further exploration.

1.6 Ethical, Social, Professional, Legal and Security Considerations

The purpose of this study is not solely to produce research findings bordering on players performance, but also to ensure that this is done professionally, and considers the societal and legal implications of conducting the research, especially as regards to the use of public datasets to ensure they do not violate GDPR regulations. It is also bounded by ethical standards which should be maintained throughout the course of the project.

1.6.1 Professional Considerations

Data integrity is a key factor in ensuring professionalism during this research. This will further strengthen the practical application of the models developed, given the connection between quality data and better model outcomes. Adequate data preprocessing and

validation will be performed in line with best practices, to ensure that the processed data is comprehensive and accurate. The findings of this dissertation will also have a meaningful effect on player scouting/drafting and performance evaluations, affecting the decision-making process of the relevant stakeholders. Ensuring that the analysis is objective and scientifically sound will be crucial to its professional application.

1.6.2 Ethical Considerations

Significant ethical questions are raised by research works, mainly in relation to bias and data privacy. Privacy needs to be a top focus because player data may contain important information. The project will ensure that personal data relating to basketball players which is used conducting the analysis and building the relevant models adheres to privacy regulations. Ethics approval has already been secured from the University so as to comply with the ethical standards and regulations governing academic research and publication.

1.6.3 Legal Considerations

This study complies with all applicable laws and data usage guidelines, guaranteeing adherence to rules controlling the use of publicly accessible datasets. Basketball-reference.com and nba.com, two reliable and openly available websites that offer unrestricted access for scholarly and non-commercial reasons, were the sources of the data used in this study. These sources have been properly credited to recognise their ownership and avoid any intellectual property rights violations. Additionally, every data processing procedure complies with the General Data Protection Regulation (GDPR), ensuring that personal data is treated intelligently and ethically throughout the study when applicable.

1.6.4 Social Considerations

By using cross-league model validation between the NBA and WNBA datasets, this study highlights the significance of equal representation in sports analytics. Sports analytics have traditionally ignored the dynamics of women's professional basketball in favour of concentrating mostly on male athletes. This study aims to give strong analytical tools that are relevant to all leagues and encourages inclusivity by incorporating WNBA statistics into the predictive modelling process. By ensuring that the methods and models created are not limited to a particular gender or league, this approach creates fairness which increases its practical relevance and applicability.

2 Literature Review

In this chapter, we will explore previous studies that have been carried out to examine the relationship between machine learning and player performance in sports, with specific focus on the NBA. The methodologies, results and significance of these studies in line with the objectives of this research will also be the purview of this chapter.

2.1 Sports Analytics and Machine Learning

2.1.1 Overview

Sports analytics has been described as the application of the Machine Learning methods and implementations to sports in order to draw useful conclusions (Bunker and Susnjak, 2022). This often significantly impact on the performance of individual athletes as well as the team either for specific games or for whole season. It can also assist teams in making predictions for upcoming talents, evaluate the market values of players and identify potential injury risks (Patrot et al., 2023). The sports industry has recently focused on big data, not only for box-score statistic cards, but also in segments, such as opponents' analysis, scouting analysis, ticket pricing, roster management, training data, health data, and social network analytics, apart from advanced performance analytics that support decision-making (Motomura et al., 2016). Sports data scientists process these qualitative and quantitative information to aid owners and technical staff with tools, providing a competitive advantage and optimizing performance, tactics and strategy (Tichy 2016).

Until recently, analytics had solely relied on simple metrics to make evaluations in the game of basketball. While they had achieved some measure of success in the past, these were limited to capturing a player's contribution or performance to the number of points, rebounds, assist or free throws made. These metrics did not project beyond the descriptive aspect of analysis and were lacking in depth and predictive power (Radovanovic, Radojicic and Savic, 2014). The emergence of data analytics technologies has transformed sports, enabling deeper insights into individual and team performance (Apostolou & Tjortjis, 2019). The adoption of complex machine learning (ML) models has integrated both simple and advanced statistics to provide better statistical analysis and uncover relationships between diverse performance factors, leading to more informed decision-making (Zuccolotto et al., 2017).

Studies have shown the continued use of machine learning algorithms in managing sports, extracting useful information from the complex data that is being generated daily. According to Lotfi and Rebbouj (2021), the nascent era of using machine learning models for prediction were only limited to simple linear regression models. But given the development of advanced data tools and technology, there is the need to transcend to more advanced algorithms like support vector machine and neural networks that can handle non-linear features. These algorithms have empowered both analysts, researchers and other stakeholders to understand both the descriptive and prescriptive nature of more advanced statistics. (Nguyen et al., 2021).

2.1.2 Significance of Player Performance Prediction in Sports

The potential benefits of tracking players performance in sports are enormous, given its vast application in player scouting, club management and developing real time strategies. For example, deep scouting knowledge and experience is required while making draft picks and transfers/trade-offs in the NBA to ensure the long-term success of the team. Recent studies have made various contributions about the significance of predicting performance. Luo and Krishnamurthy (2023) developed a deep learning model called Graph Attention Networks (GAN) with temporal convolution, to predict performance in basketball by examining the dynamic interactions of players. Similarly, a two-step prediction method combining Data Envelopment Analysis (DEA), and multivariate logistic regression was developed for predicting player performance in the NBA (Li, Wang and Li, 2019). In their research, Aditya and Priya (2024), demonstrated how data-driven methods have been developed to assess players' qualities and predict performance outcomes in football, focusing on grassroot sustainability. Other studies have also emphasized the increasing significance and prospect of performance prediction in other areas of sports, particularly in cricket (Anik et al., 2018; Devi and Sujitha Juliet, 2023), handball (Oytun et al., 2020) and hockey (Luu eta al., 2018).

2.2 Machine Learning and Deep Learning Techniques for Predicting Player Performance

2.2.1 Supervised Learning

Supervised learning techniques have remained the most widely used methods in predicting and optimizing performance in the business of sports. This method leverages the use of annotated data which are readily available datasets, especially in the NBA, to achieve its

predictive tasks. Numerous studies have applied the supervised learning approach to achieve varying results. Giannakoulas, Papageorgiou & Tjortjis, (2023) investigated how machine learning can predict the scoring performance of a player. This was achieved using three supervised learning algorithms: random forest, linear regression and multilayer perceptron. These supervised learning algorithms were applied to existing datasets and the comparative analysis of performance provided to determine the model with the best fit. Stephanos et al., (2021) developed logistic regression and the support vector regressor to determine the potential capacity of players to surpass a performance threshold over a particular season.

Decision trees have also been utilised to predict the defensive contributions and shooting efficiency of players (Yongliang and Haibin, 2023). Similarly, Wang and Fang (2021) used logistic regression and decision tree to predict the likelihood of players and teams performing optimally during the playoff season. The result showed linear regression performed well when compared with other models. However, linear models have also performed poorly when compared with random forest to predict the player efficiency ratings, and consequently identify the elite performing players in the NBA (Nguyen et al., 2021). When used to rank players according to their contributions to team success, gradient boosting algorithms like XGBoost have shown better model outcome than traditional statistical methods (Oughali, Bahloul and El Rahman, 2019). This ensemble approach has provided a new approach for the recent developments in supervised learning for player performance prediction.

Despite the effectiveness of supervised learning techniques, it is usually faced with some challenges. According to Jiang, Gradus and Rosellini, (2020), overfitting is a common problem associated with supervised learning method, especially when it is being applied to concatenated or complex datasets. This has often led to inconsistent result in the model outcomes. Moreso, supervised learning models frequently require large and high-quality datasets to train, which can be a downside for lesser leagues like the women basketball that lack the access to some advanced metrics that can be further explored for prediction

2.2.2 Unsupervised Learning

Studies have shown unsupervised learning being developed in predicting performance. Though not as prevalent as the supervised learning model, they have become relevant in recent years, given their ability to discover patterns and correlations in unlabelled datasets.

Agarwal, Sethia and Indumathy (2020) built the K-nearest neighbour (KNN) model to predict players statistics for the upcoming season. Similarly, Hidden Markov Models (HMMs) have been trained on sequence data, by using real-time game logs to predict player's decision-making during play (Papageorgiou, Sarlis and Tjortjis, 2024). Furthermore, the use of Principal Component Analysis (PCA) has been widely explored by researchers for predictive analysis (Nguyen et al, 2021; Ke, Bian and Chandra, 2024) These methods have been especially successful in identifying patterns that span seasons or leagues, like the development of three-point shooting in professional basketball.

Unsupervised learning techniques present several difficulties despite their potentials. The inability to interpret cluster assignments or latent features is a major problem that might hinder coaches' and analysts' capacity to extract useful information (Apostolou and Tjortjis, 2019). Furthermore, the features selection and data preprocessing methods have a significant impact on the quality of the results, and this may lead to bias when it is poorly executed. Another key disadvantage is the lack of labels which sometimes makes it impossible to verify the accuracy of the patterns found, requiring manual interpretation or additional supervised analysis (Nasim, Manzur and Priyanka, 2021).

2.2.3 Deep Learning Techniques

Deep learning techniques have become powerful tools, offering new perspectives in predicting players performance. Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs) have emerged as the 'twin models' of deep learning for player prediction, given their capacity to capture intricate, non-linear correlations in high-dimensional datasets, unlike supervised learning techniques. Shah and Romijnders, (2016) used sequential data to demonstrate how RNNs estimate player contributions over several games. Their result showed RNNs can reliably handle time-dependent trends by outperforming traditional regression techniques in predicting Player Efficiency Rating (PER) and game results. Similarly, Harmon et al. (2020) used CNNs to analyse tracking data, extracting temporal and geographical information to accurately forecast shot success rates—a notable improvement above feature-engineered models. These developments demonstrate how deep learning can be used to leverage rich datasets, including enhanced player statistics and spatiotemporal tracking, to generate insights that teams can use.

Other studies have focused on a hybrid approach, where deep learning models have been developed with conventional techniques, using a comparative approach. Yuan (2024) carried out a comparative study using deep neural networks and gradient boosting machines (GBMs) to predict the win share and value over replacement players (VORP). Wiik et al., (2019) integrated biometric and in-game performance data with a long short-term memory (LSTM) network to evaluate player's fatigue in relation to performance across a season. When it came to forecasting player workload and performance sustainability, the LSTM model proved to be highly reliable, offering vital evidence for roster and training management.

The continuous rise of deep learning techniques is not without its challenges. One significant limitation is that this technique require large amounts of quality data to produce accurate results, which may not be reliably available (Nguyen et al., 2021). Additionally, deep learning models are computationally costly, needing substantial resources for both hyperparameter adjustment and training. Predictions can be further distorted by bias in training data, especially in cross-league analysis where variations in player dynamics and gameplay may impact the transferability of the model (Manner, 2016).

2.3 Machine Learning Applications in the NBA

2.3.1 Predictive Modelling

The field of NBA analytics has continuously relied heavily on predictive modelling in recent times as researchers, club management and analysts are now able to forecast key performance indicators like shooting efficiency, assists, rebounds etc. A variety of machine learning models have been used to achieve this objective, with relative success being measured in various aspects of predictive modelling such as determining players' winning contributions (Hsu et al., 2018; Nguyen et al., 2021), predicting games outcome and NBA fantasy optimization (Wang, 2023; Papageorgiou, Sarlis and Tjortjis, 2024) and analysing seasonal trends of team performance (Li, Wang and Li, 2019). To develop predictive models, researchers have used a variety of data sources, including as team performance measures, college statistics, and player biometrics. Studies have identified Random Forest and Support Vector Machines (SVM) as being very effective regression techniques, extracting intricate patterns in basketball datasets (Kannan et al., 2018). Other machine learning techniques like, logistic regression and deep neural networks, have also been used

to achieve significant results in performance predictions (Wang, 2023). According to studies, traditional box scores have often performed poorly in predicting performance across all levels when compared to advanced stats (Wang & Fan, 2021). Hore and Bhattacharya (2018), demonstrated the impact of NCCA statistics and the NBA draft pick ranking as essential parameters to develop an NBA player sustainability model. This model measured the performance of NBA players over five years.

These machine learning technologies in the NBA have shown prospects for delivering valuable insights to all stakeholders in the sports. However, the effectiveness of these models is usually hampered by data bias and other deficiencies. Stephanos et al., (2021) has identified the reliance on players with high volume of minutes played as a key factor that affected their model accuracy in tracking performance across various player roles.

2.3.2 Feature Engineering

Advanced metrics have now been able to provide a better perspective on the dynamics of a player's performance through the use of machine learning. Efficiency ratings and Box Plus-Minus are now being widely adopted not just to rate the offensive abilities of a player but to underline their overall contributions to the team, while considering other factors. Nguyen et al., (2021), in their study, highlighted the impact of feature engineering in improving machine learning models for prediction by enabling analysts to extract valuable features from unprocessed data. In modern NBA, players' performance has to be fully recognized even when they are not directly scoring points by employing feature combinations such as modified Box Plus-Minus (Lackritz and Horowitz, 2020). This evaluates performance using possession statistics, which adds up each player's contributions to determine points per minute. The imperative nature of game-lag has also been investigated by modern studies, providing more clarity on how machine learning is applied to forecast games in basketball. Fatigue is considered a key variable when evaluating the probability of the team to record successive wins during both the regular season and the playoffs. (Horvat et al., 2023). Recent research has created machine learning algorithms that have produced superior results and offered insightful information for player evaluations and development by examining these sophisticated statistics.

2.3.3 Games Strategy and Decision Support

Machine learning models have also been developed to improve in-game strategy and planning. Studies have shown the effectiveness of analytics in assisting coaches create

broad-based strategies against competing teams in the league by exploiting their weaknesses, using ML models to uncover insights from play-by-play data (Bunker & Thabtah, 2019). Understanding the interactions and dynamics of play in recent times has transcended beyond traditional coaching experience, given the development of the game. There is need for big data analysis to provide scenarios and better perspective of an opposition during gameplay, based on the available information. This will ultimately increase the prospect of good decision making, therefore improving the win-loss ratio of the team. Various studies have also focused on other aspects of games strategy, notably in the area of enhanced defensive strategies as a key tool in winning championships (Patrot et al., 2023), using machine learning algorithms to determine player movements in offensive plays (Tian et al., 2020), and the effectiveness of transition plays (Daly-Grafstein and Bornn, 2021). These insights provide teams with real-time decision support, enabling them to make last-minute strategy adjustments and increase their chances of success. Coaches now approach in-game decisions differently due to the evolution of machine learning (ML) in game strategy, which extract valuable information from data.

Reinforcement learning techniques have also been applied in enhancing games strategies in basketball games. Yanal et al., (2022) proposed a reinforcement learning technique called the Q-Ball, which highlighted the correction between players decisions and games outcome. This model was built under the Deep Deterministic Policy Gradient framework, allowing coaches to influence the decision-making abilities of the key players on their roster. Chen et al., (2022), developed an RL model called ReLiable to improve tactical strategies by leveraging on existing data. This was achieved using the Convolutional Neural Network (CNN) and Transformers. These methods make use of the enormous volumes of data gathered from NBA games to offer insights into team tactics, player decision-making, and game results. Although there exist some limitations in uncovering the dynamics of basketball gameplay, machine learning techniques have the underlining capacity to ensure that the relevant stakeholders are equipped with good decision-making players and coaches construct strategies and make decisions during games.

2.4 Challenges and Ethical Considerations

2.4.1 Data Quality and Integrity

One of the fundamental issues in sports analytics is data quality, arising from data imbalance, missing data, and anomalies that usually affect the performance of these models.

Maintaining data integrity is quite important in the data mining process, especially since basketball datasets are usually merged from various sources. In their discussion of the difficulties presented by dataset differences, Zovak et al., (2019), emphasized that models are more likely to produce inaccurate predictions when they have been developed with poorly prepared datasets. It is vital to ensure that data remain consistent, and the quality is not compromised at every point.

2.4.2 Ethical and Privacy Concerns

The rise of big data in sports, especially in the NBA has called for more discussions around the need to protect the players' privacy. Ethical concerns about privacy and equitable representation have increased as player data is being tracked and examined more frequently (Arora et al., 2023). Concerns over privacy rights are raised by players' private performance metrics, particularly if predictive analysis is made public. To prevent biases, adequate analysis must also be carried out appropriately. To maintain ethical standards in sports analytics, adherence to data privacy laws and tactful treatment of public data is very important.

3 Practical Research Methodology

3.1 Data Mining Framework

This section discusses the methodologies employed and the rationale for selecting the methodology. This study aims to evaluate the performance of three machine learning models - Support Vector Regressor, Random Forest Regressor and the Recurrent Neural Network in predicting the player efficiency ratings and win share using historical data, player biodata and match statistics. To achieve this, the Cross-Industry Standard Process for Data Mining (CRISP-DM) was considered.

CRISP-DM is a popular six-step process used for the delivery of a range of machine-learning projects. The phases of CRISP-DM include:

- Data Understanding
- Data Acquisition and Understanding
- Data preparation
- Modelling
- Evaluation
- Deployment

The CRISP-DM methodology was selected due to its ubiquity across the data science industry and the flexibility of implementation - it can be implemented either using agile or waterfall project management practices (Hotz, 2024). The diagram below demonstrates the workflow of this highly effective methodology.

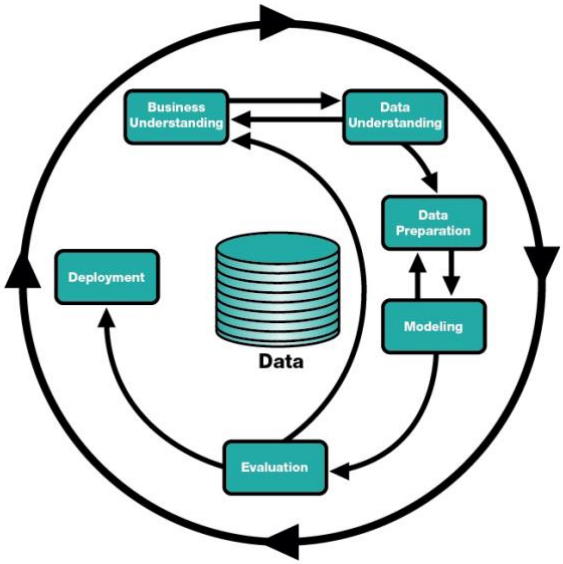


Figure 3.1: CRISP-DM Methodology (Source: Cardenas et al., 2020)

3.1.1 Business Understanding

This research aims to use machine learning to predict the Player Efficiency Ratio (PER) using historical data, player biodata and match statistics for players in the NBA and WNBA. The results from this models can be adopted by industry stakeholders in making positive decision that can improve sport business and the quality of the game.

3.1.2 Data Acquisition and Understanding

The datasets were primarily sourced from publicly available data sources containing qualitative and quantitative data about a player's biodata, match statistics and historical data across five seasons (2019/2020, 2020/2021, 2021/2022, 2022/2023, and 2023/2024). The match statistics and player advanced statistics datasets were sourced from Basketball Reference (https://www.basketball-reference.com/) while the player biodata was sourced from the NBA Advanced Stats (https://www.nba.com/stats/players/bio). The datasets were obtained using manual file export for each season.

The player statistics dataset contains the following features:

Variable	**Description**
Player	The player's name (string).
Age	The player's age during the season (string).
Team	The team the player represented during the season (string).
Pos	The player position (string)

G	The number of games the player participated in during the season (string).
GS	The number of games the player started (string).
MP	The cumulative minutes the player was on the court (integer).
FG	The total number of field goals made (integer).
FGA	The total number of field goal attempts (integer).
FG%	The percentage of field goals made by a player (float).
3P	The number of three-point shots made by a player (float).
3PA	The number of three-point shots attempted by a player (integer).
3P%	The percentage of three-point attempts made by a player (float).
2P	The number of two-point shots made by a player (integer).
2PA	The number of two-point shots attempted by a player in a team (integer).
2P%	The percentage of two-point attempts made by a player (float).
eFG%	A weighted field goal percentage that accounts for the extra value of three-pointers (float).
FT	The number of free throws made by a player (float).
FTA	The number of free throws attempted by a player (float).
FT%	The percentage of free throws made by a player (float).
ORB	The number of rebounds collected on the offensive end by a player (integer).
DRB	The number of rebounds a player collected on the defensive end (integer).
TRB	The total number of rebounds a player made (integer).
AST	Total number of assists (integer).
STL	The number of times a player took the ball away from an opponent (integer).
BLK	The number of times the player blocked an opponent's shot (integer).
TOV	The number of times the player lost possession of the ball to the opposing team (integer).
PF	The number of personal fouls committed by the player (integer).
PTS	The total number of points scored by the player (integer).

Table 1 showing the variables and descriptions from the player statistics.

The Advanced Player Statistics dataset contains the following features.

Variable	**Description**
Player	The player's name (string).

Pos	Player position
Age	The player's age during the season (string).
Tm	The team the player represented during the season (string).
G	Games played during the season (integer).
MP	Minutes played (integer).
PER	A player's performance efficiency rating for the team and season (float).
TS%	A player's shooting efficiency metric for a team and season (float).
3PAr	A player's three-point attempt rate for a team and season (float).
FTr	A player's free throw rate for a team and a season (float).
ORB%	A player's offensive rebound percentage per team per season (float).
DRB%	A player's defensive rebound percentage per team per season (float).
TRB%	A player's total rebound percentage per team per season (float).
AST%	A player's assist percentage per team per season (float).
STL%	A player's steal percentage per team per season (float).
BLK%	A player's block percentage per team per season (float).
TOV%	A player's turnover percentage per team per season (float).
USG%	A player's usage percentage per team per season (float).
OWS	A player's offensive win share per team per season (float).
DWS	A player's defensive win share per team per season (float).
WS	A player's win share per team per season (float).
WS/48	A player's offensive win share per 48 minutes for each team per season (float).
OBPM	A player's offensive box plus-minus per team per season (float).
DBPM	A player's defensive box plus-minus per team per season (float).
BPM	A player's box plus-minus per team per season (float).
VORP	A player's value over replacement player per team per season (float).

Table 2 showing the variables and descriptions from the advanced player statistics. This was sourced from http:basketball-reference.com/

The player biodata dataset contains the following features:

Variable	**Description**
Player	Player name (string)
Team	The team the player represented during the season (string).

Age	Age of the player (integer)
Height	Height of player in centimeters (float)
Weight	Weight of player in pounds (float)
College	The university a player played for before draft pick (string).
Country	Player's country of origin (string).
Season	The season the biodata is being measured (string).

Table 2 showing the variables and descriptions from the player biodata. This was sourced from http:nba.com/stats/players/bio

3.1.3 Data preparation

3.1.3.1 Concatenation & Merging

The individual files from all five seasons for each dataset were concatenated to create a unified, comprehensive dataset. This was done pandas function, pd.concat().

```
seasons = ['2023-24', '2022-23', '2021-22', '2020-21', '2019-20']
nba_datasets_ = [nba_2023_, nba_2022_, nba_2021_, nba_2020_, nba_2019_]
cols_to_remove = ['Unnamed: 24', 'Player-additional', 'Unnamed: 19', 'Rk']

combined_nba_ = pd.DataFrame()

for i, d in enumerate(nba_datasets_):
    d['Season'] = seasons[i]


    combined_nba_ = pd.concat([combined_nba_, d])
```

Figure 3.2: Python code showing concatenation of the datasets

The player statistics data and advanced statistics data were merged on the player's name, team name and season fields using an inner join. The resulting merged data was then joined to the player biodata datasets on the player's name and season using the left join.

3.1.2.1 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to have clear understanding of the dataset by examining its attributes, patterns, trends and correlations, therefore acting as the foundation for further pro-processing and the model building phase. Key Python libraries such as pandas, NumPy, matplotlib, seaborn, sklearn and keras as well as their dependencies were downloaded and imported as shown in Appendix B. The next step was to preview the

dataset to ascertain its structure and content, using the df.head() function from the pandas library. It is important to inspect the Dataframe to understand the data before proceeding to further analysis and data pre-processing. The figure 3.3 below shows the snapshot of the first 5 rows of the dataset.

```
# Previewing the dataset
data.head()
```

	Player	Pos	Age	G	MP	PER	TS%	3PAr	FTr	ORB%	...	PTS	WEIGHT	GP	Height_cm	Draft_status	Years Since Draft	PF	PG	SF	SG
0	Precious Achiuwa	C	24	25	437	15.0	0.512	0.276	0.247	12.3	...	7.7	243.0	74.0	207.264	1.0	3.0	0	0	0	0
1	Precious Achiuwa	PF	24	49	1187	14.5	0.564	0.167	0.234	13.3	...	7.6	243.0	74.0	207.264	1.0	3.0	1	0	0	0
2	Bam Adebayo	C	26	71	2416	19.8	0.576	0.041	0.381	7.4	...	19.3	255.0	71.0	210.312	1.0	6.0	0	0	0	0
3	Ochai Agbaji	SG	23	51	1003	8.1	0.531	0.570	0.080	3.9	...	5.4	215.0	78.0	198.120	1.0	1.0	0	0	0	1
4	Ochai Agbaji	SG	23	27	638	7.1	0.453	0.375	0.196	6.6	...	6.7	215.0	78.0	198.120	1.0	1.0	0	0	0	1

5 rows x 36 columns

Figure 3.3: A preview of the first 5 rows of the dataset

Next, the distribution of key features across the five seasons was examined. The total number of players across the five seasons is shown in Figure 3.4 below. It was observed that the season with the highest number of players is the 2021-2022 season with 605 players. The median age across the five seasons is 25 years. The mean weight of players was lowest in 2023-2024 with a mean weight of 214.7 pounds while the highest weight was observed in 2019-2020 with a mean weight of 217.2 pounds.

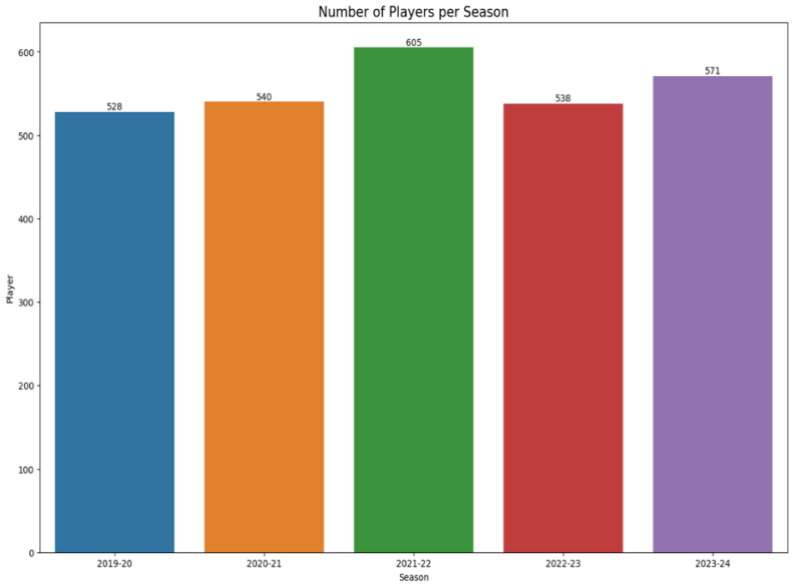


Figure 3.4: python visualization showing the number of players per season

It is important to examine the correlations among the variables to understand the relationships among variables and also to avoid issues such as multicollinearity during the modelling phase. As shown in Figure 3.4, there was a high correlation between PER, BPM, WS/48, OBPM, OWS, PTS and TS%. A high correlation relationship was also observed between OWS, DWS and WS.

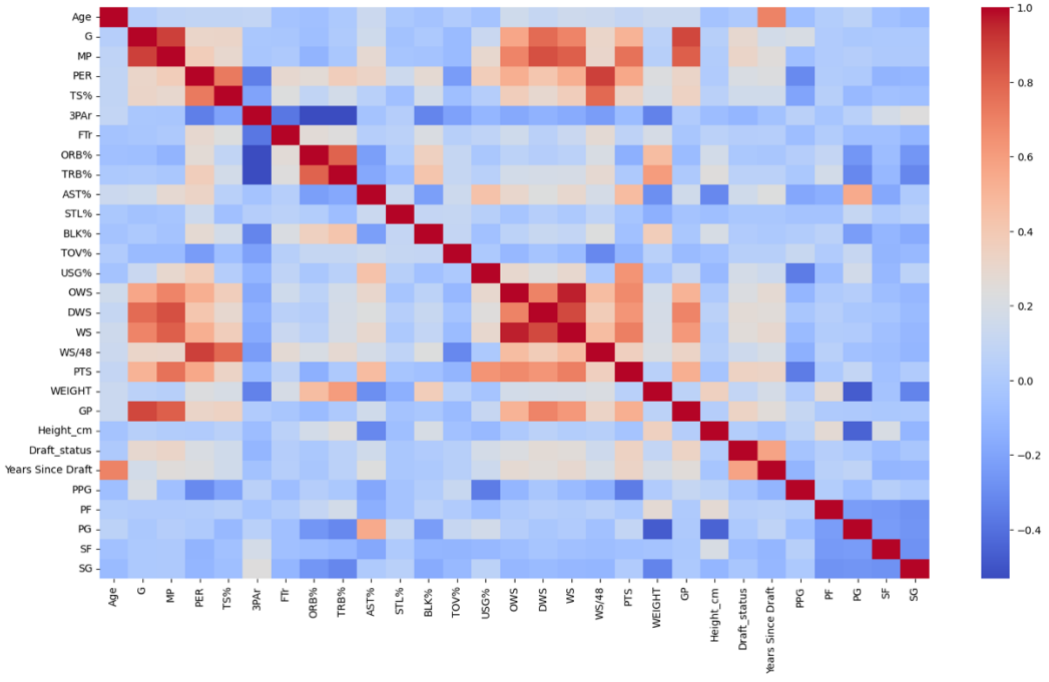


Figure 3.5: Code snippet showing correlation plot

3.1.3.2 Handling Null Values

The heatmap below as shown in Figure 3.6 visualized the missing values as present in the TS%, 3PAr, FTr, weight, college, country, draft year, draft round, games played, height_cm and draft_status fields. The missing values were resolved by dropping the rows of those variables as well as applying feature removal, based on their relevance to the target variables. This method was preferred over imputing the mean in the missing values for two reasons. One of the reasons is that the missing values do not significantly reduce the sample size of the datasets since they are within ten percent of the total number of observations. Also, including the mean values into key variables such as True Shooting (TS%) and three-points attempts (3PAr) can reduce the variability of the feature, thereby affecting data quality and the overall model performance.

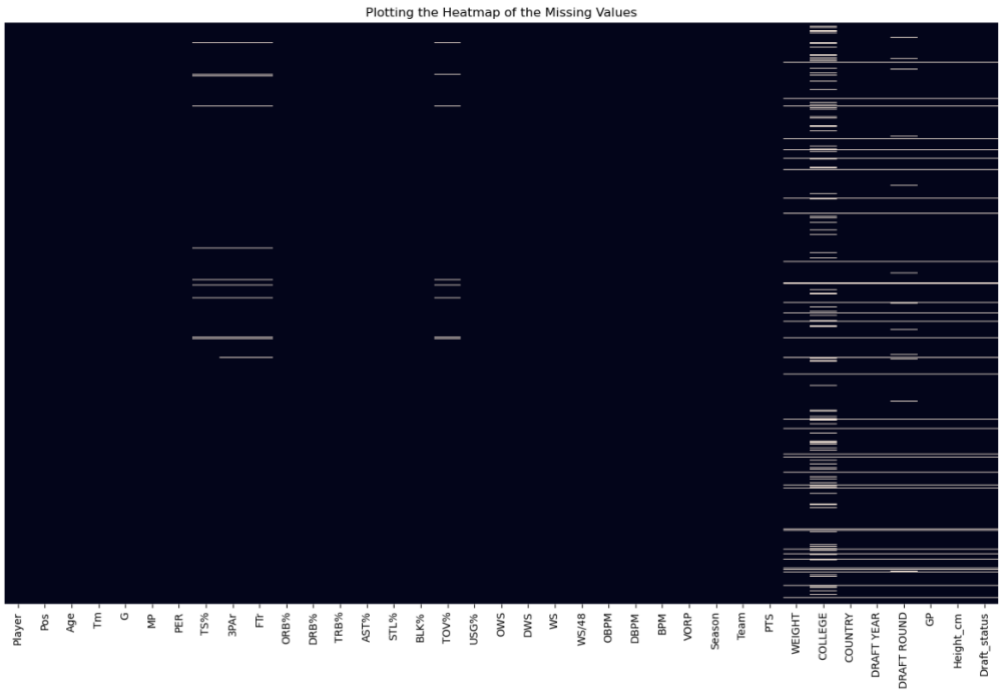


Figure 3.6: Heatmap showing missing values in the dataset

3.1.3.3 Removing Unnecessary Columns

Merged datasets often include columns or variables that are either redundant or not necessary to the overall objective of the study. One of the first steps taken was therefore the removal of unnecessary columns. The pandas library function “df.drop(columns=id)” was used to drop the “id” columns because they were not required for our analysis.

3.1.3.4 Scaling the Dataset

To ensure the datasets were in the same format, the standard scaler was used to fit the numerical features into the same scale.

```
# scaling the features
StandardScaler_model = StandardScaler()
data_one[features] = StandardScaler_model.fit_transform(data_one[features])
```

Figure 3.7: Code snippet that displays scaling the features

3.1.3.5 Splitting the Dataset

As shown in Figure 3.7, the dataset was split into train and test using the season field where the season 2019-2023 was used as the training dataset while the test data was for the 2023-2024 season.

```
1 # splitting the dataset into train and test using season
2 data_train = data_one[data_one['Season'] != '2023-24'].drop(columns='Season')
3 data_test = data_one[data_one['Season'] == '2023-24'].drop(columns='Season')
```

Figure 3.8: python code showing splitting the dataset into train and test

3.1.3.6 Reshaping the Dataset

To ensure the dataset was in the right format for the RNN model, the training dataset was reshaped to a tensor format using NumPy's reshape method.

```
1 # Building the dataset fo the RNN model
2 data_train_rnn = np.reshape(data_train.drop(columns=['PER']).values, (data_train.shape[0], 1, data_train.shape[1] - 1))
3 data_test_rnn = np.reshape(data_test.drop(columns=['PER']).values, (data_test.shape[0], 1, data_test.shape[1] - 1))
4
5 # Getting the target data in shape
6 per_train_rnn = data_train['PER'].values
7 per_test_rnn = data_test['PER'].values
```

Figure 3.9: python code showing the reshaping the train data for RNN

3.1.4 Modelling

The machine learning models built in this study were based on two key research approach.

- The Single-Task
- The Multiple-Task

3.1.4.1 The Single-Task approach

This approach is aimed at predicting the Player Efficiency Rating (PER) of players using the available NBA dataset for the periods between 2019 and 2024 regular seasons. The PER was chosen for this study because of its vast acceptability as a robust metric for assessing a player's overall contributions during gameplay. It was created by John Holinger, a sport journalist who assigned weighted points to players based on the time and situation to determine efficiency, which is then divided by the number of playing minutes (Hach et al., 2023). To measure a player's performance in relation to their pace during the game and the league's average performance level, the PER formula considers a number of factors, such as points made, assists, blocked shots, fouls, free throws, shots made, missed shots, rebounds, steals, and turnovers .

3.1.4.2 The Multi-Task Approach

This approach explored the process of simultaneously predicting two target variables in the given datasets. namely, the Player Efficiency Ratings (PER) and Win Share, using the same NBA datasets. This decision is influenced by the complementary nature of both variables in evaluating performance. Win share is an important metric in modern day basketball as it calculates a player's contributions to team wins, emphasizing both offensive and defensive inputs (Koji and Konaka, 2023). It is also considered as a key factor in the MVP selection at the end of the NBA regular season. The algorithms used to facilitate these tasks were the Support Vector Machine Regressor, Recurrent Neural Network (RNN) and Random Forest Regressor. These models were developed under the Python framework and utilized powerful python-enabled libraries such as Sklearn and TensorFlow.

3.1.4.3 Support Vector Regressor

According to Ovirianti, Zarlis and Mawengkang, (2022), the support vector regressor is a supervised machine-learning model that uses the kernel trick (linear, polynomial, radial basis function) to transform the feature space. Hence, it can be used to model non-linear relationships between input features and the target variable.

The SVR function is depicted below:

$$y = \sum_{i=1}^{N} \left(a_i - a_i^*\right) \cdot \langle x_i, x \rangle + b$$

Figure 3.9: Linear SVR Function

$$y = \sum_{i=1}^{N} (\alpha_i - \alpha_i^*) \cdot \langle \varphi(x_i), \varphi(x) \rangle + b$$

$$y = \sum_{i=1}^{N} (\alpha_i - \alpha_i^*) \cdot K(x_i, x) + b$$

Figure 3.10: Non-Linear SVR Function

```
# instantiating the models
RandomForestRegressor_model = RandomForestRegressor()
SVR_model = SVR()
```

```
# Getting the target variable
target = ['PER']

# training the models
RandomForestRegressor_model.fit(data_train.drop(target, axis=1), data_train[target])
SVR_model.fit(data_train.drop(target, axis=1), data_train[target])
```

Figure 3.11: python code showing the training of the Random Forest and SVR models

As shown in Figure 3.11 the support vector regressor was instantiated using the SVR module and the default hyper-parameters were used. The training data was then used to fit the SVR model.

3.1.4.4 Recurrent Neural Network Regressor

Recurrent Neural Networks (RNNs) are a type of neural network specifically designed for processing sequential data. The recurrent connections allow information to be passed from one step to the other (Bisong, 2019). RNNs process data iteratively, with the current input and the hidden state from the previous step both influencing the hidden state at each step. This makes the algorithm very suitable for task where sequential dependencies are important.

The general mathematical representation of an RNN is as follows:

$$h_t = f\,(W_x\, x_t + W_h h_{t-1} + b)$$

where:

- h_t : Hidden state at time t
- x_t : Input at time t
- W_x and W_h : Weight matrices for input and hidden state
- b : Bias term
- f : Activation function (represented by the 'tanh')

For this study, the RNN regressor was implemented with three SimpleRNN layers, each containing 64 neurons, created with the hyperbolic tangent as the chosen activation function. The output layer is a dense layer with one or two neurons, depending on whether the single-task or multi-task model is being trained. The model was trained using an epoch of 20. The RNN was optimized using the Adam optimizer. The Adaptive Estimation optimizer is an algorithm for optimization technique with gradient descent, which has become very popular in deep learning model building, specifically due to its efficiency. Finally, the model was evaluated using the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) metrics.

```
# Create the model
model = Sequential([
    # First RNN layer with return_sequences=True
    SimpleRNN(64, activation='tanh', return_sequences=True, input_shape=(1, data_train.shape[1] - 1)),
    SimpleRNN(64, activation='tanh', return_sequences=True),
    SimpleRNN(64, activation='tanh'),
    Dense(1)  # Output layer for regression
])

# Compile the model
model.compile(optimizer='adam', loss='mae', metrics=['mae'])  # MAE for regression

# Fitting the model
history = model.fit(data_train_rnn, per_train_rnn, epochs=20, batch_size=32, validation_split=0.2)
```

Figure 3.12: RNN Model Architecture in Python

3.1.4.5 Random Forest Regressor

The random forest algorithm is an ensemble learning method that can be used for regression and classification tasks and is built from multiple decision trees during training and outputs the mean prediction for regression from all the trees (Lyu, Wang and Zhang, 2024). Random Forest is particularly advantageous for this study due to its ability to handle non-linear relationships and high-dimensional data. Hyperparameters such as the number of trees,

maximum depth, and minimum samples per leaf were tuned during the training process to optimize model performance.

Random Forest can be mathematically expressed as

$$\hat{y} = \frac{1}{n}\sum_{i=1}^{n} T_i(x)$$

Where

y^ : Predicted value

$T_i(x)$: The prediction of the *i*-th tree for input.

n : The total number of trees.

3.1.5 Rationale for Selected Algorithms

The algorithms for this research were primarily chosen because of their ability to assist achieve the overarching aim of this study, which is to predict player performance in basketball. This practical advantage makes the Support Vector Regressor, Random Forest and Recurrent Neural Networks (RNN) well suited to implement this regression task, by applying them on historical NBA datasets.

As mentioned earlier in this chapter, SVR can be effectively used to identify non-linear relationships between variables. This method transforms input data into higher-dimensional feature spaces, allowing the algorithm to fit a linear hyperplane in that space to minimize prediction errors. SVR can be applied to NBA player performance prediction, which often involve the integration of variables that have no linear relationship and are sometimes complex to analyze. Additionally, previous studies have adopted SVR in implementing predictive modelling tasks and published results which showed that SVR outperformed linear regression in predicting player efficiency (Nguyen et al. 2021) Furthermore, SVR's kernel flexibility allows it to capture specific patterns in subsets of data, such as performance trends based on player positions or teams, making it a valuable tool for granular player analysis.

Random forest is widely seen as being popular among researchers in predicting player performance, given its ensemble technique that can be applied to both linear and non-linear interactions. Large collections of decision trees are built, trained, and their outputs are averaged for regression tasks as the algorithm carries out its predictive modelling task. Significant benefits are offered by this ensemble technique in terms of enhancing generalisation and decreasing variation. The choice of random forest is predicated on its ability to properly handle datasets that contain high dimensionality and heterogeneity, a phenomenon that is often associated with NBA datasets. Random Forest handles this effectively by partitioning data into smaller subsets and learning independently from each tree. Moreso, the feature importance scores are heavily associated with random forest and will be highly valuable in the evaluation stage of this research to analyze the relative impact of the independent variables on the target valuables in both the single and multi-task approaches. Studies such as Oughali et al. (2019) have highlighted Random Forest's superior performance in predicting advanced basketball statistics compared to traditional regression methods, further validating its inclusion in this study.

Recurrent Neural Networks (RNN) process sequential data and model temporal dependencies, which are essential in predictive modelling. This is one of the key reasons for its adoption, given the aim of the research expand its horizon by including a deep learning algorithm along with SVR and RF. This was carried to aligns this research with the increasing adoption of deep learning in sports analytics. RNNs is quite unique in its operations, given its use of recurrent connections that enable information to flow across time steps, ensuring that the patterns are well captured. This is valuable when analyzing performance in basketball per game. RNNs are particularly effective in capturing these trends, as demonstrated by Shah and Romijnders (2016), who used RNNs to estimate player contributions across multiple games and recorded relative success. Harmon et al. (2020) also built RNNs to analyze sequential tracking data in basketball, achieving high success rate in the results when predicting shooting efficiency. This capacity to model intricate temporal relationships supports the objective of this research to develop robust, time-sensitive predictive models for player performance.

The combination these algorithms ensures a comprehensive approach to predicting player performance in basketball, as the diversity in these algorithms will boost the quality of this research and reduce the bias that may be encountered from adoption a single algorithm or approach.

3.1.6 Performance Evaluation

The trained models were evaluated using the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), as expressed in the project objective.

3.1.6.1 Mean Absolute Error (MAE)

Mean Absolute Error is used to measure how close predicted values are to the actual values. It is the average of absolute errors. Mean Absolute Error (MAE) is a fundamental metric for evaluating the performance of regression models. It provides a clear and intuitive understanding of the accuracy of predictions. By calculating the average absolute difference between predicted and actual values, MAE is able to ascertain the model's error magnitude without being influenced by the direction of errors (Ahmed, 2023). This makes it an invaluable tool for assessing and comparing different regression models, aiding in the process of model selection and refinement. The MAE is expressed mathematically below.

$$\frac{1}{n}\sum_{i=1}^{n}|(DPi - DAi)|$$

Where

DPi - Predicted value

DAi - Actual value

3.1.6.2 Root Mean Squared Error (RMSE)

Root Mean Square Error, also knowns as the root mean square deviation measures the difference between values that are predicted by a model and values that are observed. It often considered the standard deviation of residuals. RMSE quantifies how dispersed these residuals are, revealing how tightly the observed data clusters around the predicted values (Hach et al., 2023). Unlike the MAE, the RMSE is usually sensitive to outliers and can potentially lead to an overestimation of the model's errors. Lower RMSE value usually indicates a good model performance. A high RMSE value shows a poorly performing model.

RMSE is generally expressed as

$$\sqrt{\frac{\sum_{l=1}^{N}(y_P - y_i)2}{N}}$$

Where

Y_p = Predicted value

y_i = Actual value

N = Number of observations

4 Analysis of Results

4.1 Model Results

4.1.1 Single-Task Approach

Given the mean of the actual value at 12.89, the results of the evaluation metrics showed that when predicting the PER, the RNN was the best-performing model with an RMSE value of 1.48. This is about 11% of the actual mean, which mean it has the least error rate when compared to Random Forest and SVR with values at 2.24 and 2.65 respectively. The MAE report also showed that the best-performing model was the RNN with an MAE value of 0.98 while the Random Forest model was the least-performing model with a value of 1.25 and 1.14 for the SVR model.

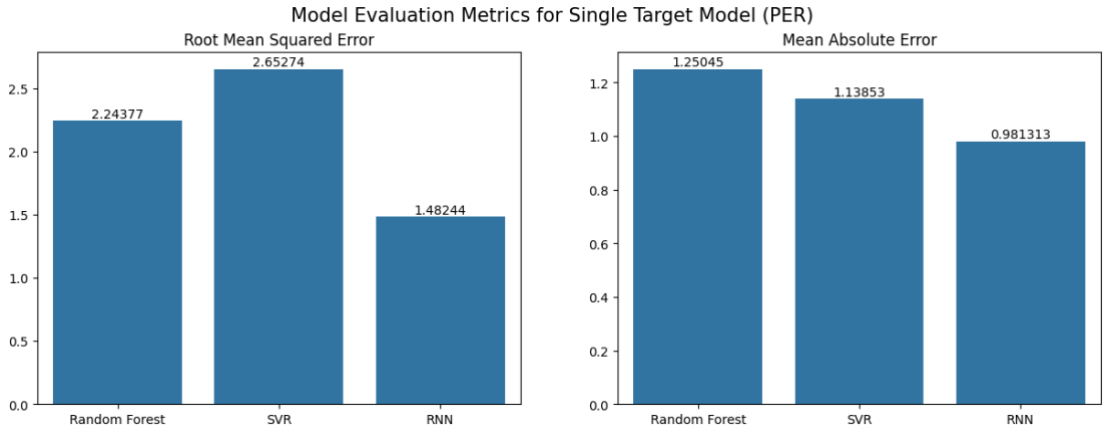


Figure 4.1: Model Evaluation Metrics of Single-Task Model

4.1.2 Multi-Task Approach

In predicting the PER, both evaluation metrics ranked the RNN as the best-performing model with a value of 1.79 and 1.13 for the RMSE and MAE respectively. The SVR model had a RMSE and MAE values of 2.71 and 1.30 while the Random Forest model was 2.29 and 1.36 respectively. Conversely, in both evaluation metrics, the RNN was the least-performing model when predicting the WS while the SVR performed best when evaluated with the RMSE and MAE**.** As shown in Figure 4.2, the RMSE values for the WS were 0.89 for random forest, 0.88 for SVR models while the RNN model was 1.23. Also, the MAE values for SVR was 0.52, while random forest and SVR both have 0.56 and 0.75 respectively.

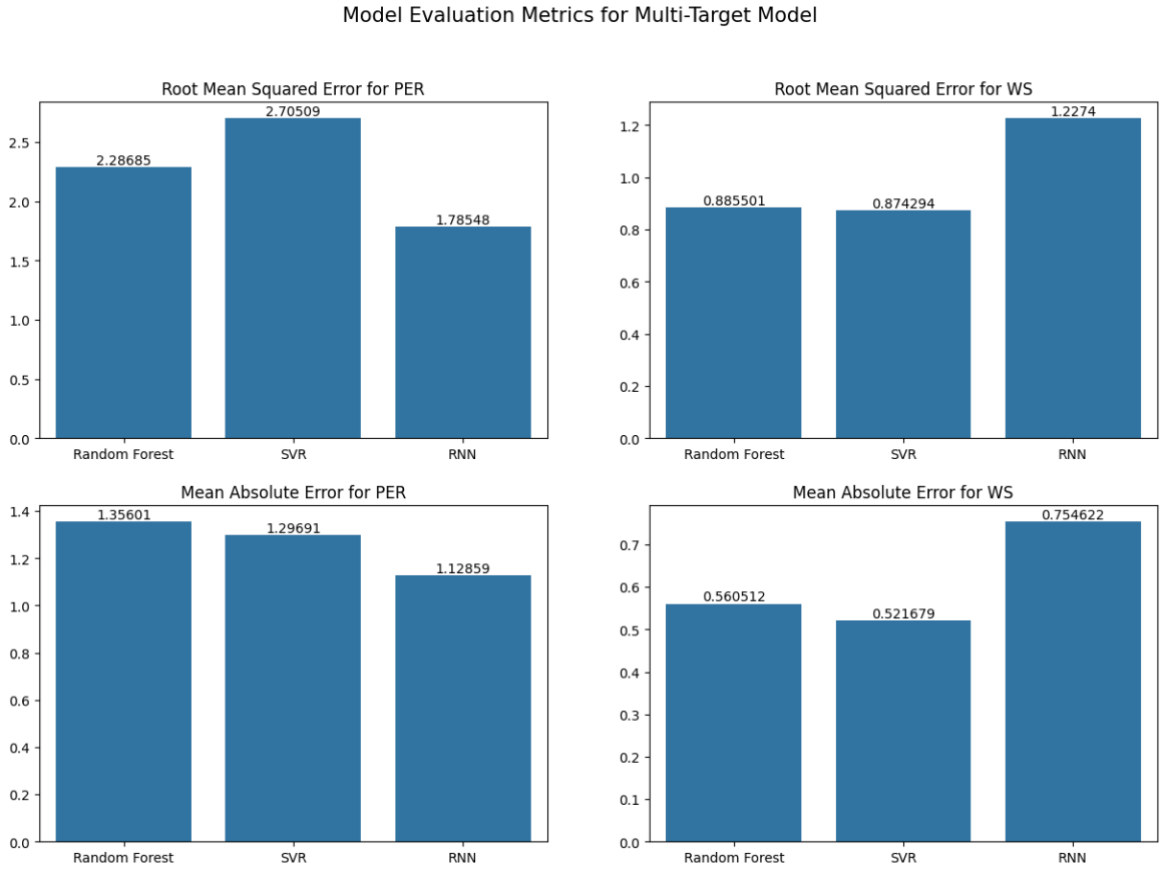


Figure 4.2: Evaluation metrics for Multi-Target Model

4.2 Discussion of Results

This section will give a comprehensive discussion of the results obtained from this study. This will attempt to give explanation to the variability in the model performance and the possible factors contributing to the performance.

4.2.1 Single-Task Approach

As shown in Figure 4.1, the RNN was the best performing model when measured with both performance metrics. This superior performance may be attributed to RNN's capacity in effectively handling sequential dependencies and complex non-linear trends, better than traditional machine learning algorithms. The model is configured to retain meaningful relationships between features, making it sitable for the dynamic nature of basketball performance metrics. The model’s consistency to outperformed the other machine learning algorithms further highlights the continued evolution of deep learning in capturing complex interactions in sports data.

The SVR model exhibited the highest error in terms of the RMSE, with a relative deviation of about 20.6% from the actual values. This poorer performance may be a function of its sensitivity to hyperparameter tuning and potential difficulty in handling complex, high-dimensional relationships. Although SVR and Random Forest provide insightful information, they need to be optimised further to match RNN's performance.

4.2.2 Multi-Task Approach

The RNN showed similar strong performances in predicting the PER under the Multi-Task approach, recording the best performance when compared with SVR and random forest. The reasons for these outcomes have been well discussed in 4.2.1. However, when predicting the Win Share (WS), SVR recorded the best performance in both RMSE and MAE when compared with the other algorithms. The SVR's strong showing for WS prediction may be due to its relative strength in working with linear relationships, which may dominate WS dynamics. On the other hand, the RNN's poor performance in predicting the Win Share in both metrics may be attributed to nature of the model, which is essentially built to handle sequential data. It may then struggle with a metric like the WS that is a cumulative metric.

The multi-task approach showed varying strengths and weaknesses as seen among the models for predicting PER and WS. While RNN is the best-performing model for PER, it struggles to make good prediction with WS. On the other hand, SVR is able to record optimal result with the WS but struggled with PER. Random Forest provides moderate performance across both metrics and tasks, showcasing its relative flexibility in prediction analysis. These outcomes underscore the importance of selecting models based on specific target characteristics and highlight the utility of multi-model approaches in complex predictive tasks.

4.3 Feature Importance

As shown in Figure 4.3, in the Single-Task model, the top five key variables in predicting the PER were True Shooting (TS%), Usage Percentage (USG%), Offensive Win Share (OWS), Total Rebound Percentage (TRB%) and Total Points (PTS). True Shooting emerged as the most important feature in predicting PER by a high margin. This conforms with apriori expectations, given that PER somewhat leans more towards offensive capabilities. True shooting is a measure of a player's ability to make accurate shots on the free throw line

and the combined field goals. A high TS% indicates a high level of efficiency, usually typical of highly skilled players who can convert scoring opportunities frequently. This is followed by the Usage Percentage (USG%) which shows a player's involvement on the court in relation to the team plays. A high USG% demonstrates key contributions made and shows that such player makes good decisions to the offensive output of the team.

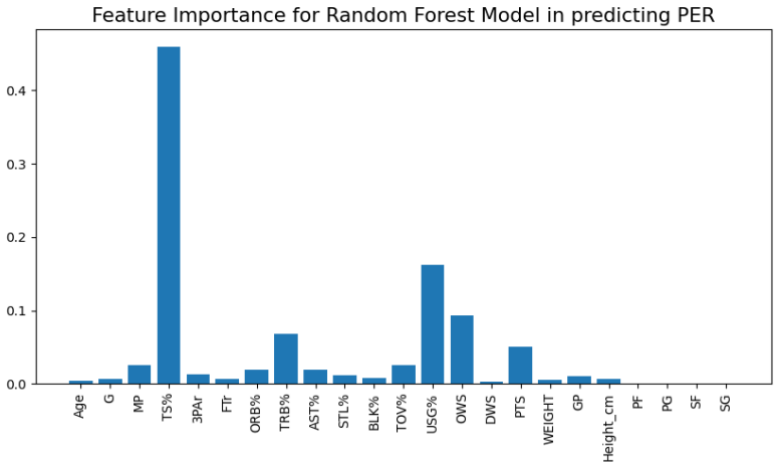


Figure 4.3: Feature importance in predicting PER using the Single Target Model

In the multi-task model, the top five features in predicting the PER were TS%, USG%, TRB%, PTS and TOV%. The top five most important features across the single and multi-target models are expected because PER is a metric computed based on a player's offensive performance. As shown in Figure 4.4, in predicting the WS, the top five features of importance were Minutes Played (MP), True Shooting (TS%), Total Points (PTS), Free Throw Rate (FTr) and Three-point attempt rate (3APr).

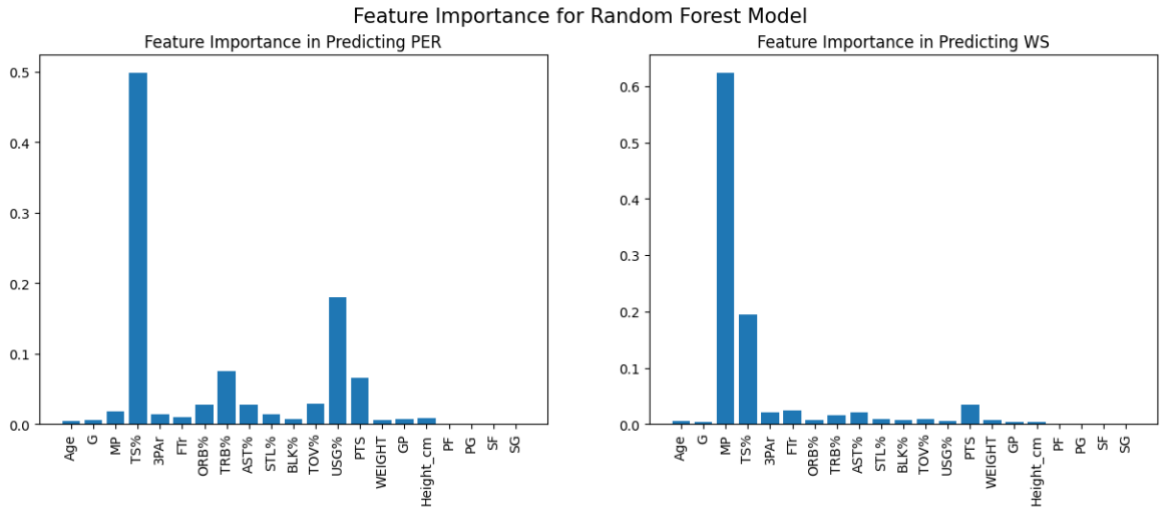


Figure 4.4: Feature importance in predicting PER and WS using the Multi-Task Model

While the same features dominated the PER in the Multi-Task model, the WS is shown to have been heavily influenced by the Minutes Played (MP). This can be explained given that more playing time gives a player more opportunities to make contributions to the team win. Therefore, the longer a player stays on the court, the more he is likely to increase his accrued points, assists, steals, rebounds etc., all of which impact the win share. Also, the coaching staff are more inclined to give more minutes played to the high performing members of the team.

4.4 Comparison of Results with Existing Literature

In this section, results obtained from this research is compared to existing literature to encapsulate the contributions of this study in the domain area as well as the general research landscape. According to Brassington (2017), RMSE and MAE are mainly differentiated by the sensitivity to outliers and, hence are often used interchangeably in evaluating model performance. The results show that when outliers are present in the data, the RMSE ranks the RNN model as the best-performing model, but MAE ranks the SVR model as it is not very sensitive to outliers. In their study, Yahyasoltani et al. (2023) utilised RSME to evaluate the predicted performance of several regression models in predicting the PER. They observed that SVR and KNN were the best-performing models with an RMSE of 0.33. Radovanović (2014) predicted the efficiency of NBA players using data from the 2011/12 season. He observed that the linear regression model performed best, with an MAE value of 0.044, compared to the SVM, Gaussian Process, and Neural Network models. The values he obtained from the best-performing model were lower than the values recorded in this research.

Seshadri (2024) compared the performance of Lasso Regression, Random Forest and Neural Network regressor (three layers consisting of two hidden layers of 128 and 64 neurons each) in predicting the PER of NBA players in the 2022-2023 season. He observed that the Random Forest Regression model was the best-performing model with an MSE value of 255.72 (RMSE of 15.99). The result is in contrast with the findings of this study. The RMSE value obtained for random forest (2.24) in this study is lower than what was observed by Seshadri (2024). This may be due to more robust data which is composed of advanced player statistics, including the player biodata. The difference in data processing techniques may also account for this disparity.

Wang et al. (2022) evaluated the performance of random forest and Feedforward Neural Networks in predicting the Win Share in the NBA using data from the 1980 season to the 2019 season. They reported Mean Square Error values of 0.014 (RMSE - 0.118) and 0.009 (RMSE - 0.0945) for the Random Forest model and the feedforward Neural Network model respectively. The lower RMSE values obtained in the study by Wang et al. (2022) as compared to this study may be due to the more abundant data points used by Wang et al. (2022), given that this research restricted the data scope to five NBA seasons.

This study explored the predictive performance of Random Forest, SVR, and RNN models for Player Efficiency Rating (PER) and Win Share (WS), providing novel insights into their strengths and weaknesses. While the RNN demonstrated strong performance for PER, its struggles with WS highlight the variability of model suitability across target variables. Comparisons with existing literature reveal the impact of data quality, preprocessing techniques, and dataset size on model performance. The findings not only validate the importance of robust data engineering but also underscore the need for tailored modelling approaches to enhance predictive accuracy in sports analytics.

4.5 Model-Testing using Cross-League Validation

The trained-RNN model was used to predict the performance of WNBA players from the 2019-2024 season, to further test and validate the model that has been developed. The RNN was chosen based on the performance shown in the modelling stage and also being the most suitable model for prediction. As shown in Table 4.1 below, the MAE and RMSE for PER were 2.09 and 3.04 respectively while the values for WS were 1.28 and 1.71 respectively.

Metrics	PER	WS
MAE	2.09	1.28
RMSE	3.04	1.72

Table 4.1: Table showing evaluation metrics for prediction using the WNBA dataset

Given the mean of the actual value for PER at 12.09 and the WS at 1.18, this implies that the model was did not show a strong performance in predicting the PER and WS of players across the entire season. Several reasons can be attributed to the poor performance of the model on the WNBA data, and the major challenge may be the existing dynamics of both

leagues. For instance, the NBA has more minutes played (48 minutes per game) as against the WNBA. Minutes played, as previously discussed under the feature importance sub-heading, is highly significant in affecting of a player's performance and contributions.

Secondly, the datasets contained in the WNBA for the same periods covered is significantly lower than the NBA. The female league currently has 12 teams (with an additional 13th in the 2025 season), compared to the NBA that comprise a total of 30 teams with 82 games in the regular season. This clearly demonstrates the disparity in the datasets. The trained RNN may have struggled to appropriately capture the best result based on the relative smaller sample size (less than 1000 observations) for the compiled seasons.

4.6 Strengths and Weaknesses Identified in the Models

This section will therefore discuss the strengths and weaknesses identified in the trained models based on the results.

Model	**Strengths**	**Weaknesses**
Support Vector Machine	Suitable for small to mid-sized datasets which is one of the rationales for its adoption in this study.	SVR are sensitive to hyperparameters (Harlianto and Setiawan, 2024).
	The kernel trick makes its very suitable in handling non-linear relationships.	The absence of feature importance in its modelling makes its difficulty to extract the relative contributions of the variables to the PER and WS.
	SVR is less sensitive to outliers when compared to the Root Mean Square Error (RMSE), given its	Sensitivity to noise and imbalanced data.

	better performance with the MAE results.	
Random Forest	Effective when applied to high-dimensional datasets which helps to mitigate against overfitting. This is main due to its ensemble characteristics.	Likely to produce a less efficient outcome when the correlation between the features is high. This is evident in the multi-task model results.
	Ability to handle non-linear relationships.	Provides a complex computational process, as it usually takes a high amount of computing power to aggregate its predictions.
	The feature importance of Random Forest is an essential one, given its capacity to extract the key predictors of player performance, which is one of the underlying objectives of this study.	Difficulty in interpretability when compared to other models.
Recurrent Neural Network	Demonstrates high proficiency in capturing sequential data, which accounts for its best performing results in predicting the PER.	Training and tuning the model can be time and resource consuming.
	Handles sequential data effectively, enabling robust	RNN usually suffers from model interpretability.

	feature learning in complex datasets .	
	Demonstrates flexibility in multi-task learning by learning shared and task-specific patterns.	High sensitivity to hyperparameters such as epoch and hidden units.

Table 4.2 showing the strengths and weaknesses of the SVR, RF and RNN based on the results

5 Project Evaluation and Reflection

5.1 Overview of Project

The overarching aim of this study is to predict player performance in the NBA using machine learning techniques. In achieving this main objective, the research approach was focused on the Single-Task Model which predicted the player efficiency ratings and Multi-Task Model, which was applied to predict both the player efficiency ratings (PER) and Win Share (WS) of NBA players. Data collection involved extracting the NBA datasets from verifiable sources, which covered a period of five seasons (2019/2020 – 2023/2024) and incorporated more advanced variables than conventional stats. This is done to ensure that the overall contributions of players are well represented and better captured in the modelling stage. The next phases of this project include detailed data pre-processing to have graphical understanding of the dataset, mitigate missing values, enable hot encoding for the player positions features, and ensure data integrity during research.

Random Forest, Support Vector Regressor (SVR), and Recurrent Neural Networks (RNN) are the three machine learning models trained and developed to predict performance using the historical data presented. The model outcomes were subjected to performance metrics such as the Mean Absolute Error (MAE) and Root Mean Square Error (RMSE), and the results were compared to one another to discover the best performing model for prediction. The project further examined the robustness of the trained-NBA model by performing a cross-league model validation using WNBA datasets. This is a novel approach with the objective being to highlight the application of the developed machine learning in dynamic basketball spaces, and how effective the model can capture nuances in other leagues apart from the NBA.

5.2 Objectives Review

The research objectives were succinctly stated in Chapter 1, and have been the guide for planning, and implementation of this study. Below is a comprehensive appraisal of each objective and its level of accomplishment:

Objective 1: Critically review existing literature on the impact of machine learning in predicting player performance in the NBA:

This objective was clearly achieved through an extensive review of state-of-the-art articles, journals and case studies in Chapter 2. The literature review examined the emergence of sports analytics in basketball, and how this has propelled the transition in the applied variables used from simple metrics to advanced statistics. The chapter equally investigated the various types of machine learning algorithms previously used by scholars in predicting player performance. This includes the supervised, unsupervised and deep learning algorithms. It highlighted the strengths and weaknesses of these algorithms, backed by empirical results. Additionally, there was a wide range of studies reviewed to discuss some of the key applications of machine learning in the NBA.

Objective 2: Systematically collect, preprocess, and explore historical NBA data to identify key trends, correlations, and statistical distributions:

The data collection process involved obtaining historical NBA datasets spanning five seasons from basketball-reference.com and nba.com. Data concatenation and merging, data cleaning, Exploratory Data Analysis (EDA) and pre-processing were duly performed under the CRISP-DM framework to assess significant insights into the datasets and prepare the data for the modelling stage.

Objective 3: Develop and implement a range of machine learning models, incorporating traditional statistics and advanced metrics for comprehensive player performance: Three machine learning algorithms—Random Forest, SVR, and RNN—were implemented to predict PER and WS. The research was structured into single-task approach focused on predicting PER, while the multi-target approach simultaneously predicted PER and WS. This objective was achieved through the application of state-of-the-art machine learning frameworks, ensuring a balanced evaluation of traditional and advanced metrics.

Objective 4: Evaluate the predictive accuracy and robustness of the models using relevant performance metrics such as Mean Absolute Error (MAE) and Root Mean Square Error (RMSE):

This objective was achieved by using the two performance evaluation metrics selected in the objective, which is the MAE and RMSE. This result and analysis were documented and fully discussed in chapter 4.

Objective 5: Perform cross-league model testing and validation of NBA models on WNBA datasets:

This objective was met as demonstrated in Chapter 4. However, cross-league validation highlighted the challenges of applying NBA-trained models to WNBA data. While the NBA-derived models performed relatively well on NBA data, their accuracy decreased sharply when tested on WNBA dataset. This underscored the dynamics of the various leagues that exists in basketball, with clear variability found in the performance metrics of both the NBA and WNBA.

Objective 6: Present a detailed report on the model outcomes, providing discussions and actionable recommendations for practical applications in sports analytics: The model results and discussions were systematically presented in Chapter 4, backed by graphical illustrations. Also, feature importance analysis was highlighted to provide the key contributors to the efficiency ratings and the win share, and comparisons with existing literature was equally made to demonstrate this study's point of convergence and divergence with recent studies. Actionable recommendations, such as using advanced metrics and customizing models for specific leagues, were highlighted to enhance the practical applicability of the research.

5.3 Evaluation of Methodology and Results

The methodology deployed in this study was based on CRISP-DM methodology in data science, covering phases including business and data understanding which involved dataset collection from nba.com and basket-reference.com as stated in subchapter 3.1. Explorative data analysis (EDA) was conducted to understand the structure and distribution of the dataset in subchapter 3.1.2.1. Data pre-processing was also performed, and the implementation of Random Forest, Support Vector Regressor (SVR) and Recurrent Neural Networks (RNN). The rationale for the selection of these algorithms was based on the literature review conducted in Chapter 2 and a detailed explanation of same was well documented in subchapter 3.1.5.

The target variables set out for this research are the player efficiency ratings (PER) and win share (WS). To achieve this, the research approach was categorized into the single task, where the PER becomes the target variable, and multitask approach where both the PER and WS become dual target variables. The models were trained on the pre-processed dataset and thereafter evaluated using the Root Mean Square Error (RMSE) and Mean Absolute Error (MAE). Lastly, the trained model was tested using a cross-league validation of the WNBA datasets to evaluate its robustness.

The results were encouraging and supported the overall objective of this research by showing the models' ability to predict the player performance in the NBA. The comparative analysis of metrics performance revealed RNN outperformed the other models when predicting PER in both RMSE and MAE. However, there were contrasting results when predicting the WS as SVR was the best performing model while RNN performed the least. The trained RNN model was then applied on the WNBA dataset to test its robustness. The model performed poorly, which can be attributed to several reasons, including missing variables, limited sample size from the WNBA datasets as well as league-specific dynamics. Future research should address these limitations by exploring diverse datasets, increasing the scope of the periods, focusing on cross-league model development and testing.

5.4 Personal Reflection

Working on this project over the past few months has been enlightening for me, as it has provided me with an opportunity to put into practical use the skills and knowledge that has been garnered during my study. It has given me a thorough grasp of how machine learning and statistical methods may combine with sports analytics to yield insightful results. Beyond the acquisition of technical skills, this research has pushed and challenged me to use my critical thinking skills, challenging presumptions and creating new research approaches to suit the objectives of the study.

In the early stages of the project, I understood the importance of thorough data sourcing and collection. Working with the NBA and WNBA datasets across various seasons, performing data cleaning and pre-processing improved my technical ability to handle and properly conduct data mining. Exploring through these datasets and uncovering various insights, I became proficient in data concatenation, scaling and normalization, resolving missing values, detecting anomalies and generally making the data fit and ready for model building. I have been greatly exposed to the practical workings of machine learning models as this project has provided hands-on experience with an array of machine learning algorithms such as Random Forest, Support Vector Regression, and Recurrent Neural Networks (RNNs). I learned to fine-tune hyperparameters, handle overfitting, and evaluate models using metrics such as RMSE and MAE. Implementing a multi-task prediction approach further enhanced my understanding of complex regression problems. Working extensively with Python libraries such as TensorFlow, Scikit-learn, and Pandas greatly improved my coding skills. I also gained experience in visualizing data and results using Matplotlib and Seaborn,

enabling effective communication of results findings. I learned to appreciate the essence of utilizing performance metrics like MAE and RMSE to ensure that the models developed are robust enough.

Additionally, this project has deepened my appreciation for interdisciplinary research. Bridging the gap between sports analytics and machine learning required not only technical skills but also an understanding of basketball's strategic and statistical nuances. This intersectionality enriched my perspective, encouraging me to approach problems holistically by integrating domain knowledge with computational techniques.

Apart from technical skills acquired, the project has greatly enabled me to hone various aspects of soft skills like critical thinking, time management, communication skills and problem-solving. The success of this project was therefore hinged on these skills, underscored by the importance of collaboration and seeking feedback with my supervisor. The regular consultations with my supervisor provided valuable guidance and new perspectives that enriched the research and guaranteed its success. This experience taught me that no project exists in isolation and that leveraging the expertise of others can lead to more robust and impactful results.

This project required me to think critically to navigate through challenges such as missing data, cross-league model testing and other underlying issues that were encountered in the process. Developing solutions to these problems enhanced my ability to think critically and approach issues methodically. Also, balancing the demands of a multi-phase project taught me to prioritize tasks effectively. Breaking down complex objectives into manageable milestones ensured steady progress and timely completion. Moreso, my communication skills improved massively through writing progress reports, presenting findings, and justifying methodologies to my supervisor. All these refined my ability to articulate complex technical concepts clearly and concisely.

Lastly, one of the key lessons I have learned is the importance of data integrity and quality. The ethical perspective of research was very evident from the conceptualization of the project topic to the implementation and evaluation of the models. I learnt much about data governance during this journey and this realization drove me to invest significant time in preprocessing, feature selection, and ensuring that the datasets used were both comprehensive and consistent across seasons.

Amidst all of these, I have also had to deal with several challenges during this dissertation. One of the major challenges was the process of training the Recurrent Neural Network, a deep learning model that can is complex and subject to several iterations to achieve optimal results. Managing these demands with the limited timeline, coupled with the need to satisfy other aspects of the project was rather daunting, and one that required proper planning and time management. Another challenge was also dealing with data limitation especially during cross-league model testing. There were missing advanced metrics in the WNBA datasets, such as Box Plus-Minus, Value Over Replacement Player etc. Ensuring that models trained on NBA data could perform well on WNBA data was a recurring difficulty. This necessitated adjusting the initial model configuration and re-training the model to ensure that the model validation is performed.

In conclusion, this study has been a rewarding experience that has shaped me as a person and a researcher. The lessons I have learnt about resilience, teamwork, and critical thinking will help me tackle obstacles in any field, and the technical and soft skills I have gained will surely be valuable in my future endeavours. Notwithstanding the challenges, the research confirmed my enthusiasm for data science and its revolutionary potentials to address pressing issues in the real world. The insights gained will not only inform my future work but also inspire me to continue exploring the intersection of machine learning and sports analytics with renewed curiosity and determination.

5.4.1 Project Management

This research adopted the hybrid project management which combines both the Agile and Waterfall methods to achieve its results. This approach was implemented, given the iterative process involved in this research, based on the regular consultation and feedback from the supervisor. The Agile approach is equally evident based on the methodological way research is conducted and executed. It entails proper documentation of each process, which is a Waterfall technique.

Being a Certified Scrum Master also bolstered the practical knowledge needed to complete a project of this scope within the time frame. The nature of the project therefore necessitated the rationale for adopting the hybrid of approach in this research.

5.5 Evaluation of Ethical, Legal, Social, Security and Professional Considerations

Ethical Considerations:

The objective to uphold the ethical aspects of this project was duly achieved at every phase, especially during data preparation. Due consultation was made with the project supervisor who provided collaboration and support to obtain an Ethics Approval from the university. This approval is conventionally required to begin the research process and strict adherence to data privacy. Player's anonymity was ensured during the modelling stage as the 'Player' variable was one of the columns dropped before training the models. All these measures gave the guarantee that the research was conducted in line with the stipulated university guidelines with respect to research.

Legal Considerations:

The research gave considerations to the data sources and only accessed these datasets from public domains and websites. Moreso, these sources have been repeatedly acknowledged in the course of this research, to ensure that no infringement is caused. The practical methodology aspect of this research duly conveyed this credit to the sources and also clearly stated and referenced the algorithms adopted. Great effort was put in to uphold the legal integrity of this research.

Social Considerations:

The social considerations of this project were fully addressed. With basketball being a sport, which is essentially seen as a social activity, the result outcome from this predictive task can assist stakeholders make good choices when evaluating players in terms of playing time, scouting and development. This has significant social implications. Having the right models to measure players performance in basketball can ensure that all players are given fair assessment on the team roster, during scouting or when negotiating contracts. Moreso, the inclusion of the WNBA datasets to test the trained models highlighted the dynamics and challenges of female basketball, raising research awareness and potential support for not only basketball, but also women sports in general.

Professional Considerations:

The CRISP-DM processes of data collection, data preprocessing, modelling and evaluation carried in the most professional manner, to enhance the credibility of the final results. Basketball is a profesional sport, and the research in this domain area needed the adherence to industry best practices. The hybrid approach adopted in the project management of this research provided credence to the research, ensuring that each phases was completed as due and regular iterations from feedbacks were implemented as well. The models for this research were also built in the ambit of the regulations, and the results clearly presented to assist stakeholders take actions that can improve the game.

6 Conclusion and Recommendations

6.1 Conclusion

The primary objective of this study was to investigate how machine learning models can predict player performance in the NBA, proxied by the player efficiency ratings (PER) and win share (WS) as target variables, under a single-task and multi-task approach. To achieve this, the research adopted the CRISP-DM research methodology which involved understanding the project objectives, data preparation and pre-processing, modelling and evaluation. Machine learning techniques-Random Forest and Support Vector Regressor, combined with Recurrent Neural Networks (RNN), a deep learning technique, were therefore trained and applied on the pre-processed datasets, followed by a comparative analysis of their results and performances. The details of this outcomes were comprehensively documented in the previous chapters, demonstrating that the objective set out were achieved.

Findings from the research revealed the RNN emerging as the most effective model in predicting the PER under the single-task approach, outperforming random forest and SVR when measured with the performance evaluation metrics. However, while the RNN equally relatively revealed a superior outcome in predicting PER in the multi-task approach, it failed to effectively estimate the win share as a target variable, being the least performing model when compared to the other algorithms. The variability in the RNN model's performance was also evident when it was applied on the WNBA data to further test the validity and robustness of the model. By using WNBA datasets for cross-league analysis, these findings offer a novel contribution while remaining consistent with existing literature.

The study encountered several challenges at various phases. This was particularly evident during cross-league model testing, where there were some missing variables in the WNBA datasets. This necessitated the adoption of alternative solutions such that required a recalibration of the NBA data to enhance consistency in the input features and validate the model. Moreso, the evolving nature of the WNBA meant its dataset is significantly lower than that of well-established NBA, which affected the performance of the model testing as RNN model are prone to overfitting when the data size is relatively small. This has spurred the need for actionable decisions to develop other basketball leagues, especially women's basketball.

Despite these challenges, the study has made its contribution in the domain area of sport analytics by investigating the dynamics of predicting performance across leagues in basketball, thereby expanding the scope of player evaluation. Future studies can leverage on this groundwork to explore alternative modelling techniques, broaden and diversify the datasets, and further enhance cross-league analytics frameworks. Also, data mining approach that focused on incorporating advanced basketball features paved the way for the integration of advanced metrics. The findings have implications for team management, scouting, and player development, offering actionable recommendations for enhancing decision-making processes.

6.2 Recommendations

This research proposes the following recommendations, based on its findings.

6.2.1 Inclusivity and Accessibility

Giving high priority to the growth of sport analytics across all spectrums of basketball will engender inclusion among player regardless of the tier they belong to. Much attention has often been placed on the elite leagues like the NBA and this has often created a wide disparity in the level of investments. This study established the potential of using machine learning to predict player performance across leagues, highlighting the importance of tailoring metrics and models to reflect the unique dynamics of each league. Metrics that take into consideration the physiological aspects of the sport and strategic differences should be investigated by researchers. In order to promote fair decision-making procedures in scouting, coaching, and player evaluation, stakeholders should make sure that models can be adjusted to different player traits and league dynamics.

Ensuring that leagues in the lower rung of the tier can equally have good coverage and have unfettered access to data, will greatly enhance resource allocation and ultimately improve the game. Future initiatives should concentrate on cascading these technologies, developing open-source platforms, and streamlining data processing methods, as this will increase the scope of accessibility. Additionally, programs that teach athletes, coaches, and analysts how to use data-driven insights can improve the comprehension and application of these technologies at all levels of the sport. A more competitive and inclusive environment can result from making sure that everyone in the basketball ecosystem can benefit from machine learning.

6.2.2 Practical Applications

This research presents an opportunity for stakeholders in professional basketball to leverage on the research findings, facilitated by the robust methodology, to improve various aspect of the sport, such as player scouting and in-game strategy. Front office management and analysts can utilize the models developed in this study by implementing regression analysis during player trade-offs and NBA draft picks. The ability to identify potential key players that can add sustained value to the team success is dependent on efficient player performance evaluation system. This is also important for teams to navigate through the complex nature of player contracts by assessing the value of such player before making informed decisions. Additionally, coaches can become more efficient with the team roster by alloting more playing time to key players with high efficiency ratings and winning contributions, while also keeping the team healthy.

Beyond team management, this research can significantly contribute to the broader domain of sports analytics. These models can be deployed in tracking performance in platforms such as in fan engagement platforms, such as fantasy basketball leagues, where fans engage and showcase their ability to evaluate player performance from a virtual standpoint. The scalability of the methodology ensures that the framework can be adapted to other sports or leagues, fostering more data-driven approaches in sports management. To guarantee its applicability, stakeholders should place a high priority on the ongoing improvement of these models by introducing more sophisticated metrics and investigating cross-league comparisons.

6.2.3 Further Research

There is ample for further research in the domain of sport analytics, especially in the NBA. Achieving better and more robust results in the predictive modelling of player performance should be the focus of researchers who want to explore this field of study. Metrics such as spatial tracking data, shot quality, and defensive positioning, which were beyond the scope of this study, could provide deeper insights into player performance. Additionally, other key components of the sport can be infused into the model to strengthen the model outcomes. For instance, a player's injury history can provide more insights into his efficiency ratings, a factor that was not considered in this research. Investigating how some of these contextual variables can be integrated into advanced metrics could provide a more better understanding of player contributions over a period of time. Researchers should also consider further

exploration into building deep learning architectures as the neural network model utilized in this research reported a relatively good outcome in predicting player efficiency ratings.

6.2.4 Alternative Approaches

Reinforcement learning algorithms, such as Deep Q-Learning, could be used to model game situations and improve in-game decisions. These methods have demonstrated potential for improving player performance and in-game strategy. Transfer Learning is another alternate strategy that can be used to improve cross-league validation. NBA-trained models can also be modified for usage in the WNBA using Transfer Learning, which resolves the difficulties in cross-league validation. By adapting previously trained models to fresh datasets, this method lessens the requirement for intensive retraining.

The adoption of an alternate approach like unsupervised learning can be utilized to uncover hidden patterns and relationship in basketball analysis. Techniques such as clustering can be used to categorized players based on similar performances. These insights could complement the use of supervised learning models, enhancing their overall performance.

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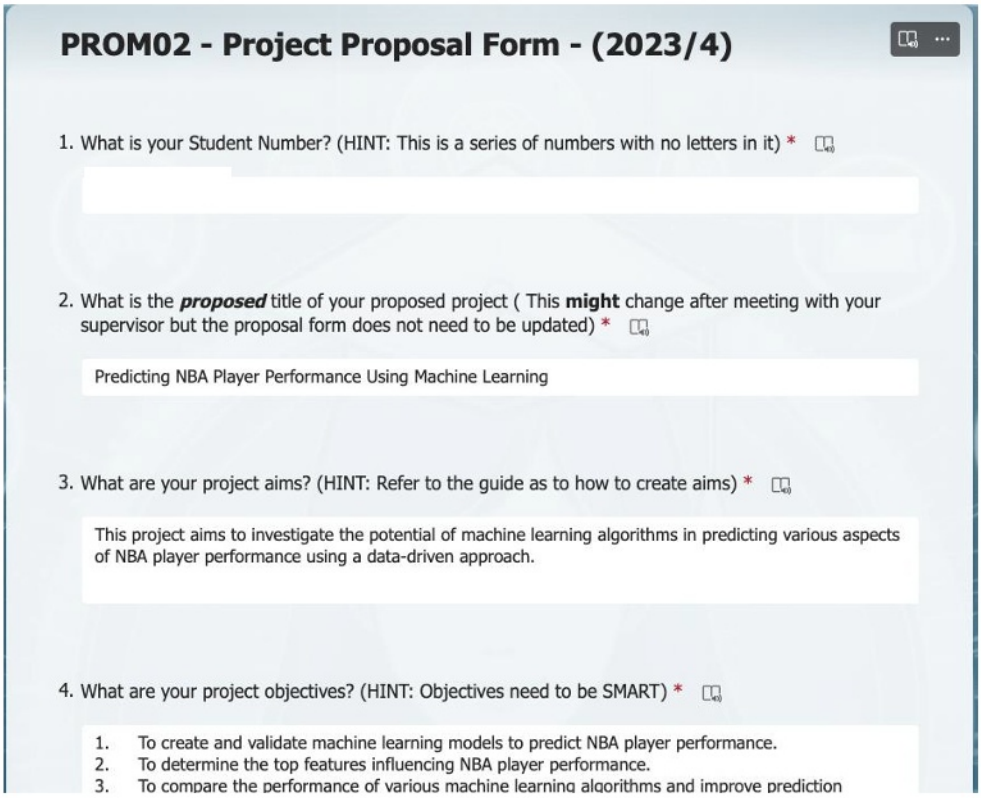
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Appendix A. Project Proposal



PROM02 - Project Proposal Form - (2023/4)

1. What is your Student Number? (HINT: This is a series of numbers with no letters in it) *

2. What is the ***proposed*** title of your proposed project (This **might** change after meeting with your supervisor but the proposal form does not need to be updated) *

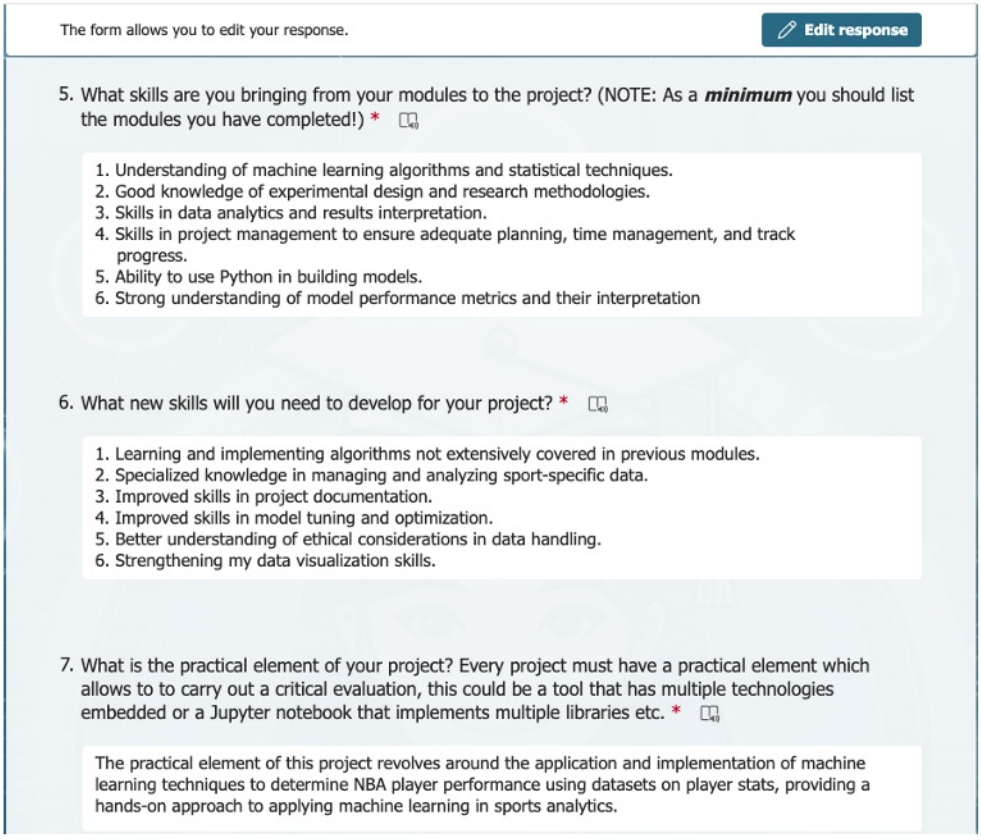
Predicting NBA Player Performance Using Machine Learning

3. What are your project aims? (HINT: Refer to the guide as to how to create aims) *

This project aims to investigate the potential of machine learning algorithms in predicting various aspects of NBA player performance using a data-driven approach.

4. What are your project objectives? (HINT: Objectives need to be SMART) *

1. To create and validate machine learning models to predict NBA player performance.
2. To determine the top features influencing NBA player performance.
3. To compare the performance of various machine learning algorithms and improve prediction



The form allows you to edit your response. Edit response

5. What skills are you bringing from your modules to the project? (NOTE: As a ***minimum*** you should list the modules you have completed!) *

1. Understanding of machine learning algorithms and statistical techniques.
2. Good knowledge of experimental design and research methodologies.
3. Skills in data analytics and results interpretation.
4. Skills in project management to ensure adequate planning, time management, and track progress.
5. Ability to use Python in building models.
6. Strong understanding of model performance metrics and their interpretation

6. What new skills will you need to develop for your project? *

1. Learning and implementing algorithms not extensively covered in previous modules.
2. Specialized knowledge in managing and analyzing sport-specific data.
3. Improved skills in project documentation.
4. Improved skills in model tuning and optimization.
5. Better understanding of ethical considerations in data handling.
6. Strengthening my data visualization skills.

7. What is the practical element of your project? Every project must have a practical element which allows to to carry out a critical evaluation, this could be a tool that has multiple technologies embedded or a Jupyter notebook that implements multiple libraries etc. *

The practical element of this project revolves around the application and implementation of machine learning techniques to determine NBA player performance using datasets on player stats, providing a hands-on approach to applying machine learning in sports analytics.

8. Please confirm that you will be applying for ethics approval. *

- (●) Yes
- ( ) No

9. Please list about 5 papers/articles/journals that you have looked at in your area of interest. (HINT: Look for papers that are current and demonstrate you have carried out some preliminary research into the area) *

Hsu, P.-H., Galsanbadam, S., Yang, J.-S. and Yang, C.-Y. (2018). Evaluating Machine Learning Varieties for NBA Players' Winning Contribution. [online] IEEE Xplore. doi:https://doi.org/10.1109/ICSSE.2018.8520017.

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Appendix B. Code Snippets

Appendix B1: Importing the Python Libraries

```
# Loading in the datasets
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split

# Visualisation
import matplotlib.pyplot as plt
import seaborn as sns

# machine learning models
from sklearn.ensemble import RandomForestRegressor
from sklearn.svm import SVR
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import SimpleRNN, Dense, Dropout

# Model management
from sklearn import metrics
from sklearn.preprocessing import StandardScaler
from sklearn.multioutput import MultiOutputRegressor

%matplotlib inline
```

Appendix B2: Merging the NBA datasets

```
seasons = ['2023-24', '2022-23', '2021-22', '2020-21', '2019-20']
nba_datasets_ = [nba_2023_, nba_2022_, nba_2021_, nba_2020_, nba_2019_]
cols_to_remove = ['Unnamed: 24', 'Player-additional', 'Unnamed: 19', 'Rk']

combined_nba_ = pd.DataFrame()

for i, d in enumerate(nba_datasets_):
    d['Season'] = seasons[i]


    combined_nba_ = pd.concat([combined_nba_, d])
```

```
seasons = ['2023-24', '2022-23', '2021-22', '2020-21', '2019-20']
nba_datasets = [nba_2023, nba_2022, nba_2021, nba_2020, nba_2019]
cols_to_remove = ['Unnamed: 24', 'Player-additional', 'Unnamed: 19', 'Rk']

combined_nba = pd.DataFrame()

for i, d in enumerate(nba_datasets):
    d['Season'] = seasons[i]


    combined_nba = pd.concat([combined_nba, d.drop(columns=cols_to_remove)])
```

Appendix B3: Explorative Data Analysis

```
[2] # Loading in the dataset
    data = pd.read_csv('NBA_Modelling_New.csv')
    wnba_data = pd.read_csv('WNBA_Modelling_New.csv')
```

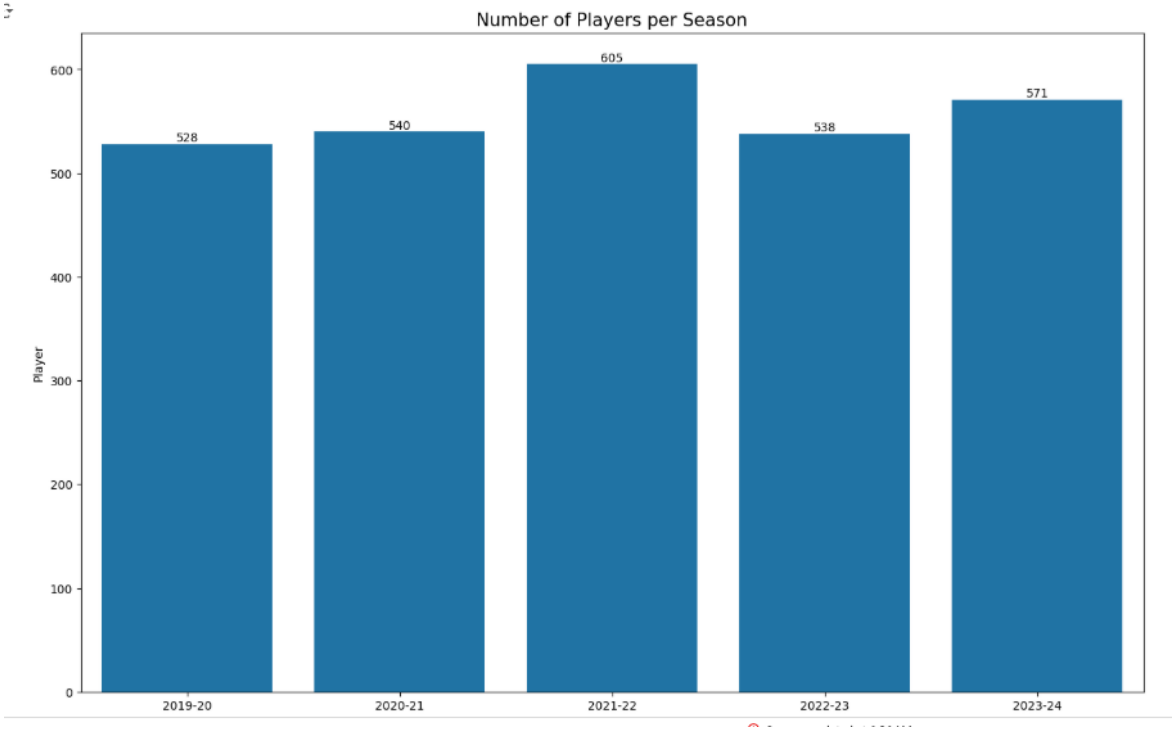
```
# Previewing the dataset
data.head()
```

	Player	Pos	Age	G	MP	PER	TS%	3PAr	FTr	ORB%	...	PTS	WEIGHT	GP	Height_cm	Draft_status	Years Since Draft	PF	PG	SF	SG
0	Precious Achiuwa	C	24	25	437	15.0	0.512	0.276	0.247	12.3	...	7.7	243.0	74.0	207.264	1.0	3.0	0	0	0	0
1	Precious Achiuwa	PF	24	49	1187	14.5	0.564	0.167	0.234	13.3	...	7.6	243.0	74.0	207.264	1.0	3.0	1	0	0	0
2	Bam Adebayo	C	26	71	2416	19.8	0.576	0.041	0.381	7.4	...	19.3	255.0	71.0	210.312	1.0	6.0	0	0	0	0
3	Ochai Agbaji	SG	23	51	1003	8.1	0.531	0.570	0.080	3.9	...	5.4	215.0	78.0	198.120	1.0	1.0	0	0	0	1
4	Ochai Agbaji	SG	23	27	638	7.1	0.453	0.375	0.196	6.6	...	6.7	215.0	78.0	198.120	1.0	1.0	0	0	0	1

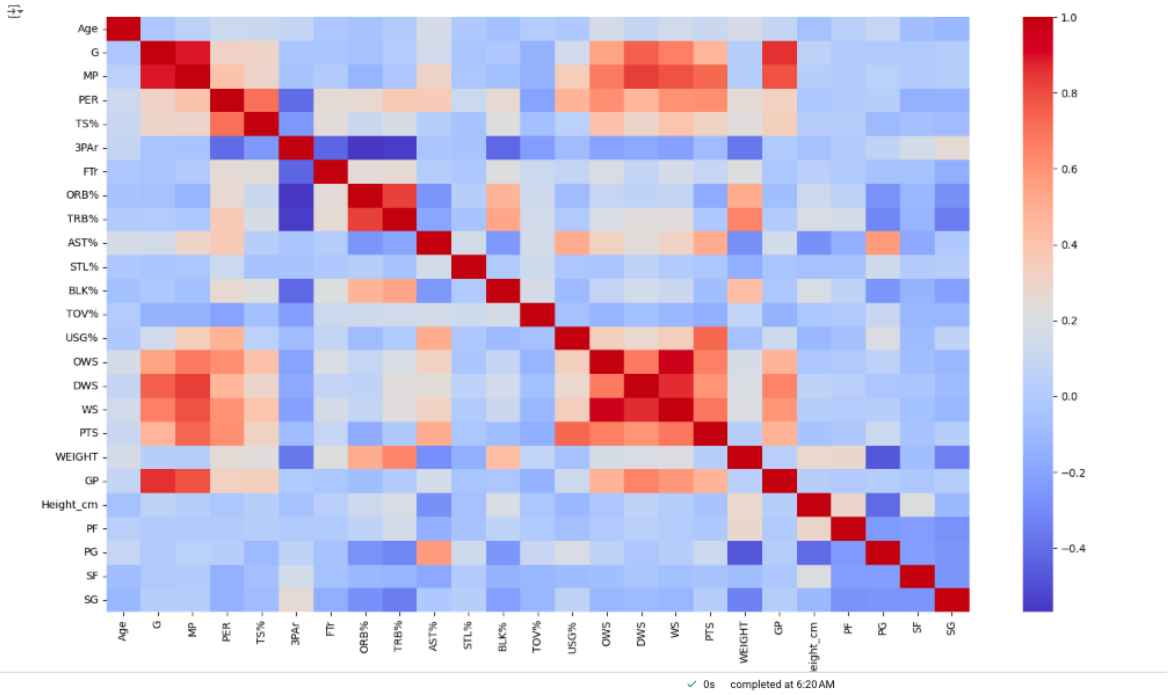
5 rows x 36 columns

```
2] # Computing the Number of Players per season
   number_of_players_per_season = data.groupby('Season')['Player'].nunique().reset_index()
```

```
plt.figure(figsize=(16, 10))
ax = sns.barplot(x='Season', y='Player', data=number_of_players_per_season)
plt.title('Number of Players per Season', size=15)
ax.bar_label(ax.containers[0]);
```



```
# Checking for correlation between features
plt.figure(figsize=(18, 10))
corr_matrix = data.select_dtypes(exclude='object').corr()
sns.heatmap(corr_matrix, cmap='coolwarm');
```



✓ 0s completed at 6:20 AM

Appendix B4: Building the Random Forest and SVR Models

```
[12] # Getting the features
     features = data.drop(columns = ['PER', 'Season', 'WS']).keys()

     # Getting a copy of the dataset
     data_one = data.copy()

     # scaling the features
     StandardScaler_model = StandardScaler()
     data_one[features] = StandardScaler_model.fit_transform(data_one[features])
```

```
[13] # splitting the dataset into train and test using season
     data_train = data_one[data_one['Season'] != '2023-24'].drop(columns=['Season', 'WS'])
     data_test = data_one[data_one['Season'] == '2023-24'].drop(columns=['Season', 'WS'])
```

```
# instantiating the models
RandomForestRegressor_model = RandomForestRegressor()
SVR_model = SVR()
```

```
[15] # Getting the target variable
     target = ['PER']

     # training the models
     RandomForestRegressor_model.fit(data_train.drop(target, axis=1), data_train[target])
     SVR_model.fit(data_train.drop(target, axis=1), data_train[target])
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/base.py:1389: DataConversionWarning: A column-vector y was passed when a 1d array was exp
  return fit_method(estimator, *args, **kwargs)
/usr/local/lib/python3.10/dist-packages/sklearn/utils/validation.py:1408: DataConversionWarning: A column-vector y was passed when a 1d a
  y = column_or_1d(y, warn=True)
▾ SVR
SVR()
```

Appendix B5: Building the RNN Model

```
[17] # Building the dataset fo the RNN model
     data_train_rnn = np.reshape(data_train.drop(columns=['PER']).values, (data_train.shape[0], 1, data_train.shape[1] - 1))
     data_test_rnn = np.reshape(data_test.drop(columns=['PER']).values, (data_test.shape[0], 1, data_test.shape[1] - 1))

     # Getting the target data in shape
     per_train_rnn = data_train['PER'].values
     per_test_rnn = data_test['PER'].values
```

```
# Create the model
model = Sequential([
    # First RNN layer with return_sequences=True
    SimpleRNN(64, activation='tanh', return_sequences=True, input_shape=(1, data_train.shape[1] - 1)),
    SimpleRNN(64, activation='tanh', return_sequences=True),
    SimpleRNN(64, activation='tanh'),
    Dense(1)  # Output layer for regression
])

# Compile the model
model.compile(optimizer='adam', loss='mse', metrics=['mse'])

# Fitting the model
history = model.fit(data_train_rnn, per_train_rnn, epochs=20, batch_size=32, validation_split=0.2)
```

```
/usr/local/lib/python3.10/dist-packages/keras/src/layers/rnn/rnn.py:204: UserWarning: Do not pass an `input_shape`/`input_dim` a
  super().__init__(**kwargs)
Epoch 1/20
46/46 ━━━━━━━━━━━━━━━━━━━━ 4s 13ms/step - loss: 217.3728 - mse: 217.3728 - val_loss: 150.6818 - val_mse: 150.6818
Epoch 2/20
46/46 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - loss: 113.6906 - mse: 113.6906 - val_loss: 25.1831 - val_mse: 25.1831
Epoch 3/20
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 25.1536 - mse: 25.1536 - val_loss: 12.4193 - val_mse: 12.4193
Epoch 4/20
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 14.2595 - mse: 14.2595 - val_loss: 9.9090 - val_mse: 9.9090
Epoch 5/20
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 12.4179 - mse: 12.4179 - val_loss: 8.6064 - val_mse: 8.6064
Epoch 6/20
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 9.6707 - mse: 9.6707 - val_loss: 7.0967 - val_mse: 7.0967
Epoch 7/20
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 7.9631 - mse: 7.9631 - val_loss: 6.2279 - val_mse: 6.2279
Epoch 8/20
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 16.5284 - mse: 16.5284 - val_loss: 5.3577 - val_mse: 5.3577
Epoch 9/20
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 8.0781 - mse: 8.0781 - val_loss: 4.9890 - val_mse: 4.9890
```

Appendix B6: Model Performance for Single-Task Approach

- Root Mean Square Error
- Mean Absolute Error

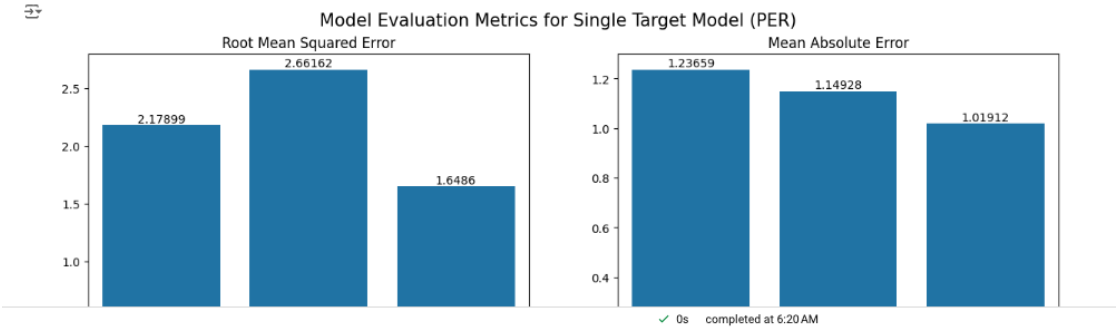
```
[20] rmse_randomforest = np.sqrt(metrics.mean_squared_error(data_test['PER'], random_forest_pred))
     rmse_svr = np.sqrt(metrics.mean_squared_error(data_test['PER'], SVR_pred))
     rmse_rnn = np.sqrt(metrics.mean_squared_error(per_test_rnn, RNN_pred))

     mae_randomforest = metrics.mean_absolute_error(data_test['PER'], random_forest_pred)
     mae_svr = metrics.mean_absolute_error(data_test['PER'], SVR_pred)
     mae_rnn = metrics.mean_absolute_error(per_test_rnn, RNN_pred)
```

```
[21] # Saving the results
     results = {'Random Forest': rmse_randomforest, 'SVR': rmse_svr, 'RNN': rmse_rnn}

     plt.figure(figsize=(15, 5))
     plt.suptitle('Model Evaluation Metrics for Single Target Model (PER)', size=15)
     plt.subplot(1, 2, 1)
     ax = sns.barplot(x=list(results.keys()), y=list(results.values()))
     ax.bar_label(ax.containers[0])
     plt.title('Root Mean Squared Error');

     plt.subplot(1, 2, 2)
     ax = sns.barplot(x=list(results.keys()), y=[mae_randomforest, mae_svr, mae_rnn])
     ax.bar_label(ax.containers[0])
     plt.title('Mean Absolute Error');
```



✓ 0s completed at 6:20 AM

Appendix B7: Training the models for the Multi-Task Approach

```
# Saving a copy of the dataset to variable
data_two = data.copy()

# Getting the multi output target
targets = ['PER', 'WS']

# Getting the features for the multi-target approach
features_two = data_two.drop(columns = ['PER',  'WS', 'Season', 'OWS', 'DWS']).keys()

# scaling the features
data_two[features_two] = StandardScaler_model.fit_transform(data_two[features_two])

# splitting the dataset into train and test using season
data_train_2 = data_two[data_two['Season'] != '2023-24'].drop(columns=['Season', 'OWS', 'DWS'])
data_test_2 = data_two[data_two['Season'] == '2023-24'].drop(columns=['Season', 'OWS', 'DWS'])
```

```
[23] # Wrapping the model with a multi regressor output
MultiForest_model = MultiOutputRegressor(RandomForestRegressor())
MultiSVR_model = MultiOutputRegressor(SVR())

# Fitting the data to the multi output model
MultiForest_model.fit(data_train_2.drop(targets, axis=1), data_train_2[targets])
MultiSVR_model.fit(data_train_2.drop(targets, axis=1), data_train_2[targets])
```

```
MultiOutputRegressor
  estimator: SVR
    SVR
```

```
[24] plt.figure(figsize=(15, 5))
plt.suptitle('Feature Importance for Random Forest Model', size=15)

plt.subplot(1, 2, 1)
plt.title('Feature Importance in Predicting PER')
plt.bar(features_two, MultiForest_model.estimators_[0].feature_importances_)
plt.xticks(rotation=90)

plt.subplot(1, 2, 2)
plt.title('Feature Importance in Predicting WS')
plt.bar(features_two, MultiForest_model.estimators_[1].feature_importances_)
```

```
# Building the dataset fo the RNN model
data_train_2_rnn = np.reshape(data_train_2.drop(columns=targets).values, (data_train_2.shape[0], 1, data_train_2.shape[1] - 2))
data_test_2_rnn = np.reshape(data_test_2.drop(columns=targets).values, (data_test_2.shape[0], 1, data_test_2.shape[1] - 2))

# Getting the target data in shape
per_train_2_rnn = data_train_2[targets].values
per_test_2_rnn = data_test_2[targets].values

# Create the model
MultiRNN_model = Sequential([
    # First RNN layer with return_sequences=True
    SimpleRNN(64, activation='tanh', return_sequences=True, input_shape=(1, data_train_2.shape[1] - 2)),
    SimpleRNN(64, activation='tanh', return_sequences=True),
    SimpleRNN(64, activation='tanh'),
    Dense(2)  # Output layer for regression
])

# Compile the model
MultiRNN_model.compile(optimizer='adam', loss='mse', metrics=['mse'])  # MSE for regression

# Fitting the model
MultiRNN_history = MultiRNN_model.fit(data_train_2_rnn, per_train_2_rnn, epochs=20, batch_size=32, validation_split=0.2)
```

```
Epoch 1/20
/usr/local/lib/python3.10/dist-packages/keras/src/layers/rnn/rnn.py:204: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When u
  super().__init__(**kwargs)
46/46 ━━━━━━━━━━━━━━━━━━━━ 4s 13ms/step - loss: 104.8756 - mse: 104.8756 - val_loss: 72.3167 - val_mse: 72.3167
Epoch 2/20
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 54.4792 - mse: 54.4792 - val_loss: 11.8103 - val_mse: 11.8103
Epoch 3/20
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 10.1862 - mse: 10.1862 - val_loss: 7.2946 - val_mse: 7.2946
Epoch 4/20
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - loss: 9.4285 - mse: 9.4285 - val_loss: 5.8135 - val_mse: 5.8135
Epoch 5/20
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 6.6414 - mse: 6.6414 - val_loss: 5.0091 - val_mse: 5.0091
Epoch 6/20
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 6.1706 - mse: 6.1706 - val_loss: 4.3215 - val_mse: 4.3215
Epoch 7/20
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 7.5212 - mse: 7.5212 - val_loss: 3.7691 - val_mse: 3.7691
Epoch 8/20
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - loss: 6.1230 - mse: 6.1230 - val_loss: 3.3833 - val_mse: 3.3833
Epoch 9/20
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - loss: 6.3282 - mse: 6.3282 - val_loss: 2.9810 - val_mse: 2.9810
Epoch 10/20
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - loss: 3.5441 - mse: 3.5441 - val_loss: 2.6790 - val_mse: 2.6790
Epoch 11/20
46/46 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - loss: 3.2675 - mse: 3.2675 - val_loss: 2.4938 - val_mse: 2.4938
Epoch 12/20
```

Appendix B8: Performance Evaluation for the Multi-Task Approach

```
[27] rmse_randomforest_per = np.sqrt(metrics.mean_squared_error(data_test_2[targets[0]], random_forest_pred_2[:, 0]))
     rmse_svr_per = np.sqrt(metrics.mean_squared_error(data_test_2[targets[0]], SVR_pred_2[:, 0]))
     rmse_rnn_per = np.sqrt(metrics.mean_squared_error(per_test_2_rnn[:, 0], RNN_pred_2[:, 0]))

     rmse_randomforest_ws = np.sqrt(metrics.mean_squared_error(data_test_2[targets[1]], random_forest_pred_2[:, 1]))
     rmse_svr_ws = np.sqrt(metrics.mean_squared_error(data_test_2[targets[1]], SVR_pred_2[:, 1]))
     rmse_rnn_ws = np.sqrt(metrics.mean_squared_error(per_test_2_rnn[:, 1], RNN_pred_2[:, 1]))
```

```
[28] mae_randomforest_per = metrics.mean_absolute_error(data_test_2[targets[0]], random_forest_pred_2[:, 0])
     mae_svr_per = metrics.mean_absolute_error(data_test_2[targets[0]], SVR_pred_2[:, 0])
     mae_rnn_per = metrics.mean_absolute_error(per_test_2_rnn[:, 0], RNN_pred_2[:, 0])

     mae_randomforest_ws = metrics.mean_absolute_error(data_test_2[targets[1]], random_forest_pred_2[:, 1])
     mae_svr_ws = metrics.mean_absolute_error(data_test_2[targets[1]], SVR_pred_2[:, 1])
     mae_rnn_ws = metrics.mean_absolute_error(per_test_2_rnn[:, 1], RNN_pred_2[:, 1])
```

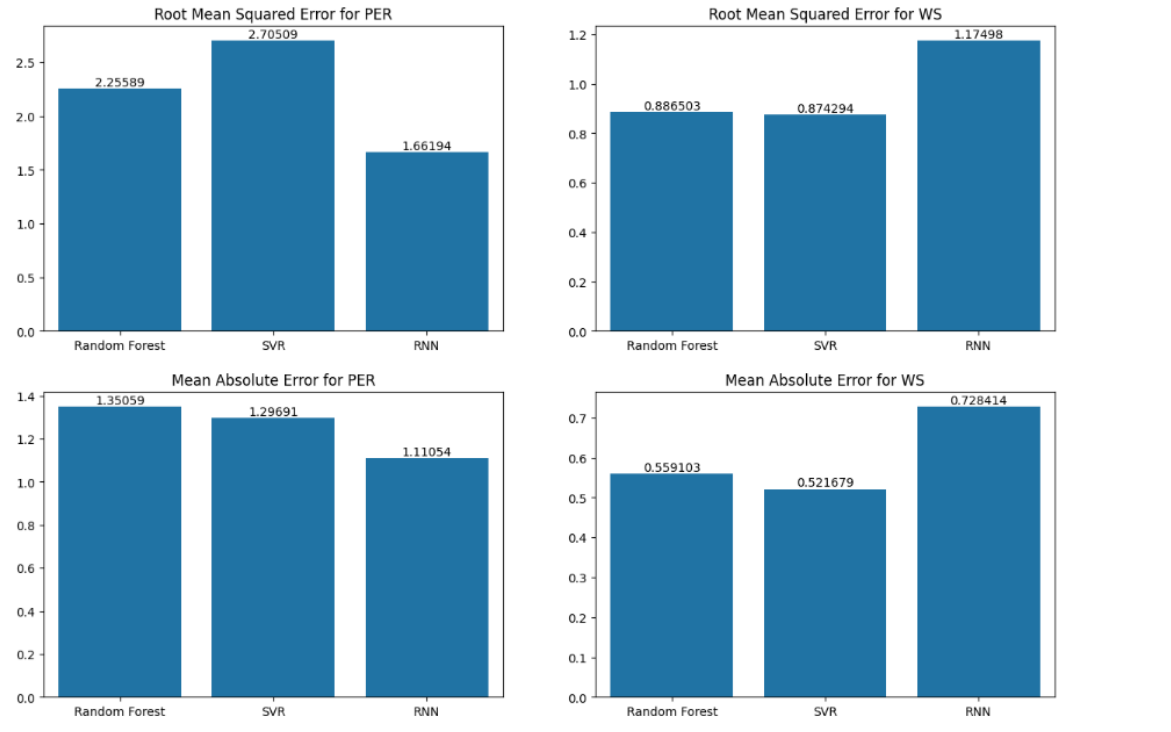
```
# Saving the results.
results = {'Random Forest': rmse_randomforest, 'SVR': rmse_svr, 'RNN': rmse_rnn}

plt.figure(figsize=(15, 10))
plt.suptitle('Model Evaluation Metrics for Multi-Target Model', size=15)
plt.subplot(2, 2, 1)
ax = sns.barplot(x=list(results.keys()), y=[rmse_randomforest_per, rmse_svr_per, rmse_rnn_per])
ax.bar_label(ax.containers[0])
plt.title('Root Mean Squared Error for PER');

plt.subplot(2, 2, 2)
ax = sns.barplot(x=list(results.keys()), y=[rmse_randomforest_ws, rmse_svr_ws, rmse_rnn_ws])
ax.bar_label(ax.containers[0])
plt.title('Root Mean Squared Error for WS');

plt.subplot(2, 2, 3)
ax = sns.barplot(x=list(results.keys()), y=[mae_randomforest_per, mae_svr_per, mae_rnn_per])
ax.bar_label(ax.containers[0])
plt.title('Mean Absolute Error for PER');

plt.subplot(2, 2, 4)
ax = sns.barplot(x=list(results.keys()), y=[mae_randomforest_ws, mae_svr_ws, mae_rnn_ws])
ax.bar_label(ax.containers[0])
plt.title('Mean Absolute Error for WS');
```



Appendix B9: Testing the RNN Model on the WNBA datasets

```
[30]  # Removing the null values
      wnba_data[features] = wnba_data[features].replace('-', np.nan)

      # Removing null values
      wnba_data.dropna(inplace=True)
```

```
      # Preparing the dataset
      wnba_copy_1 = pd.DataFrame(StandardScaler_model.fit_transform(wnba_data[features]), columns = features)
      wnba_copy_2 = pd.DataFrame(StandardScaler_model.fit_transform(wnba_data[features_two]), columns = features_two)

      # Preparing the dataset for the single target model
      wnba_rnn_1 = np.reshape(wnba_copy_1.values, (wnba_copy_1.shape[0], 1, wnba_copy_1.shape[1]))
      wnba_rnn_target_1 = wnba_data['PER'].values

      # Preparing the dataset for the multi-target model
      wnba_rnn_2 = np.reshape(wnba_copy_2.values, (wnba_copy_2.shape[0], 1, wnba_copy_2.shape[1]))
      wnba_rnn_target_2 = wnba_data['PER'].values
```

```
[32] # Predict using the WNBA dataset
     wnba_pred_1 = model.predict(wnba_rnn_1)
     wnba_pred_2 = MultiRNN_model.predict(wnba_rnn_2)
```

```
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step
```

```
[33] # Evaluation metrics
     wnba_mae = metrics.mean_absolute_error(wnba_rnn_target_1, wnba_pred_1)
     wnba_rmse = np.sqrt(metrics.mean_squared_error(wnba_rnn_target_1, wnba_pred_1))
```

```
      plt.figure(figsize=(10, 5))
      plt.title('Metrics for Prediction using WNBA', size=15)
      ax = sns.barplot(x = ['MAE', 'RMSE'], y=[wnba_mae, wnba_rmse]);
      ax.bar_label(ax.containers[0]);
```